
Chapter 6

Frequency Response of Amplifiers

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Outline

- **General Donsideration**
 - **Poles and Zeros**
 - **Association of Poles with Nodes**
- **Common-Source stage**
- **Source Followers**
- **Common-Gate stage**
- **Cascode Stage**
- **Differential Pair**



Why considering frequency response?

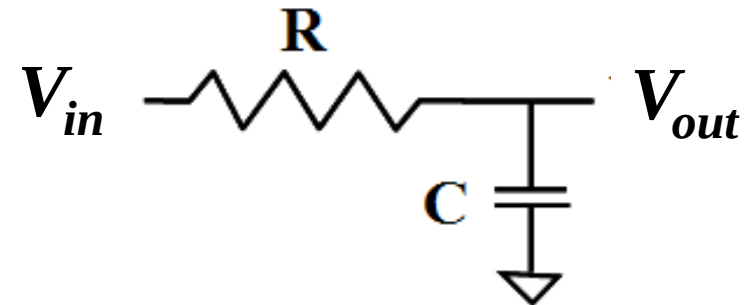
- So far, we discuss only the DC or low-frequency characteristics
 - Without considering parasitic or load capacitors
- Analog circuits must process high frequency signals
 - ADC/DAC, Filer, RF.....
- Circuit performances vary sharply at high frequencies
 - Maybe cease to work at some frequency limit
- This chapter is to talk about the frequency response limitations

Time domain analysis of a series RC circuit

- Derive the step response of a RC circuit

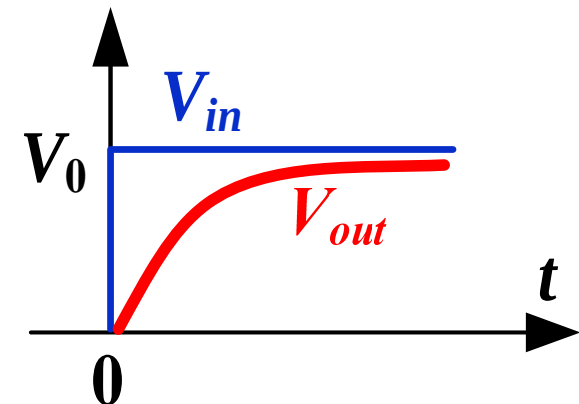
- The differential equation

$$\frac{V_{in} - V_{out}}{R} = C \frac{dV_{out}}{dt}$$



- Considering the initial condition, the solution is

$$V_{out} = V_0 (1 - e^{-t/RC})$$



- RC is called time constant

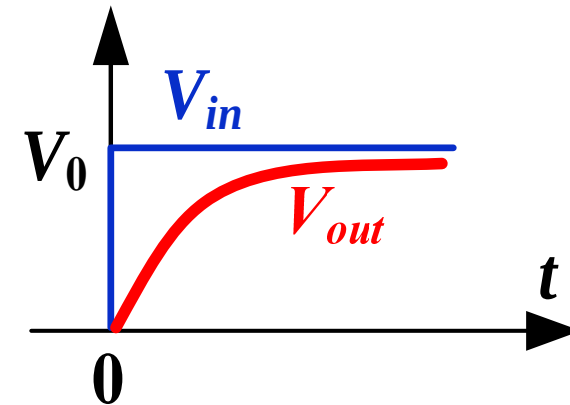
- The larger RC, the longer settling time



Time domain analysis of a series RC circuit

- The larger RC, the longer settling time

$$V_{out} = V_0 (1 - e^{-t/RC})$$



- If we apply a sinusoid signal $V_0 \sin(\omega_0 t)$ to the circuit in the steady state, what is the response at the output?
- Even more complex case: if the sinusoid signal is applied at $t=0$, what is the output response?



Frequency response analysis

- Can we describe the frequency response directly?
 - Frequency domain **transfer function**
 - Laplace transform of the **impulse response**
 - Difficult for large-scale circuits!
- Fortunately, we can obtain the transfer function from the complex impedance of device directly
- Complex impedance of a capacitor is

$$Z_C = \frac{1}{sC}$$

-
- **s** is complex frequency

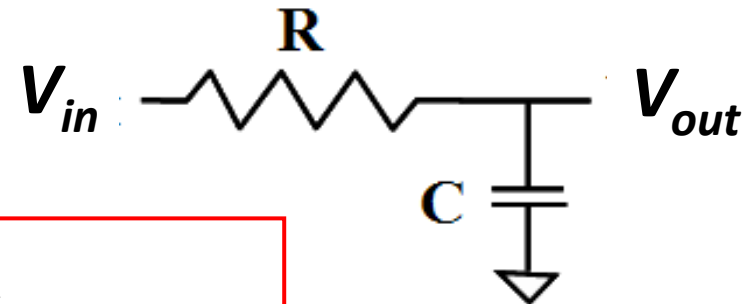
$$s = \delta + j\omega$$

- where δ is the exponential decay constant (rad /s)
- ω is the sinusoid angular frequency (rad/s)
- **Sinusoid steady state**
 - A special case, input voltage is a pure sine signal
 - No exponential decay
$$\delta = 0 \quad s = j\omega$$
 - Used to obtain the frequency response of circuits

Transfer function of a series RC circuit

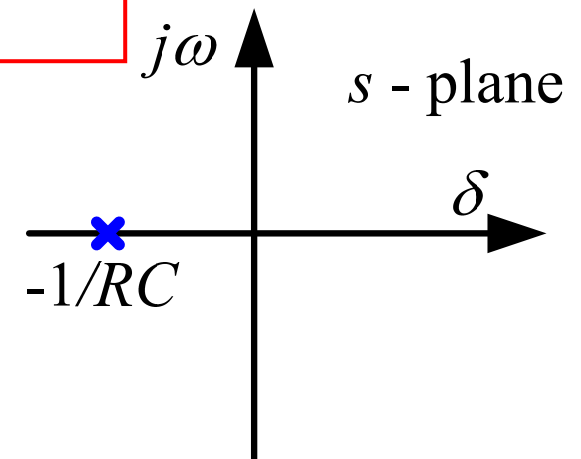
$$H(s) = \frac{V_{out}}{V_{in}}(s) = \frac{1/(sC)}{R + 1/(sC)} = \frac{1}{1 + sRC} = \frac{1}{1 + s/\omega_p}$$

where, $\omega_p = \frac{1}{RC}$



Let $D = 1 + sRC = 0 \Rightarrow s_p = -\frac{1}{RC} = -\omega_p$

s_p is defined as the **pole** of the circuit
We call it a “single-pole” function
or first-order transfer function



Frequency response of the circuit

$$H(j\omega) = H(s) \Big|_{s=j\omega} = \frac{1}{1 + j\omega RC} = \frac{1}{1 + j(\omega / \omega_p)}$$

$$\text{Amplitude: } |H(j\omega)| = \frac{1}{\sqrt{1 + (\omega / \omega_p)^2}} = \frac{\omega_p}{\sqrt{\omega_p^2 + \omega^2}}$$

$$\text{Phase: } \angle H(j\omega) = \arctan\left(-\frac{\omega}{\omega_p}\right)$$

As frequency increases, the amplitude decreases and the phase angle increases

Graphic explanation

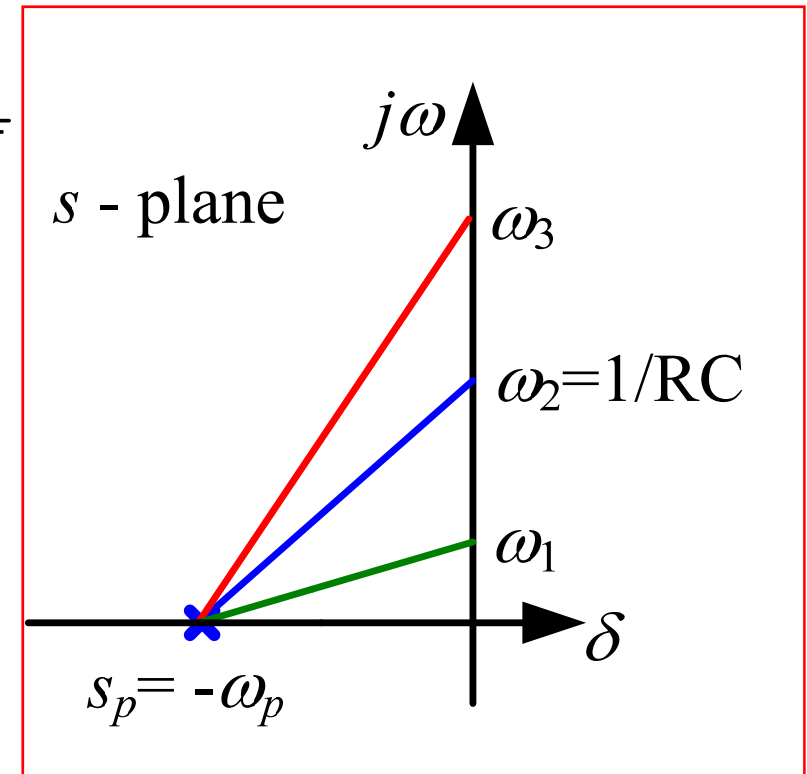
$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega / \omega_p)^2}} = \frac{\omega_p}{\sqrt{\omega_p^2 + \omega^2}}$$

$$\angle H(j\omega) = \arctan\left(-\frac{\omega}{\omega_p}\right)$$

When $\omega = |\omega_p|$,

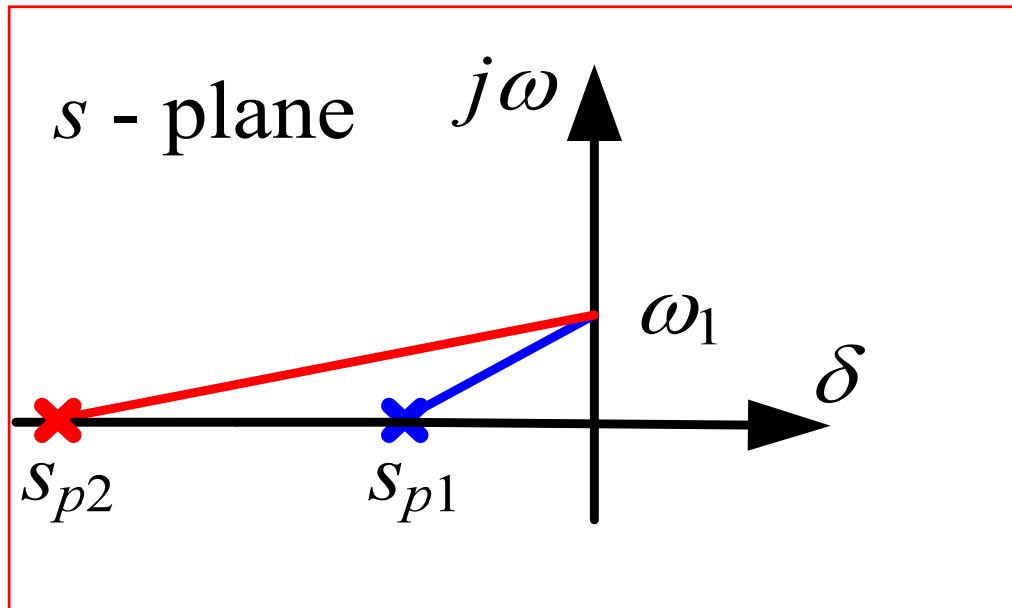
$$|H(j\omega)|_{\omega=\omega_p} = \sqrt{2} / 2 = -3 \text{ dB}$$

$$\angle H(j\omega) \Big|_{\omega=\omega_p} = -\pi / 4$$



We call this the -3dB bandwidth, or cutoff frequency

Circuit with two real poles



$$\frac{V_{out}}{V_{in}}(s) = A_1 A_2 \frac{1}{1 + s / s_{p1}} \cdot \frac{1}{1 + s / s_{p2}}$$

If $|s_{p1}| \ll |s_{p2}|$, s_{p1} is called the dominant pole



Parallel RC circuit

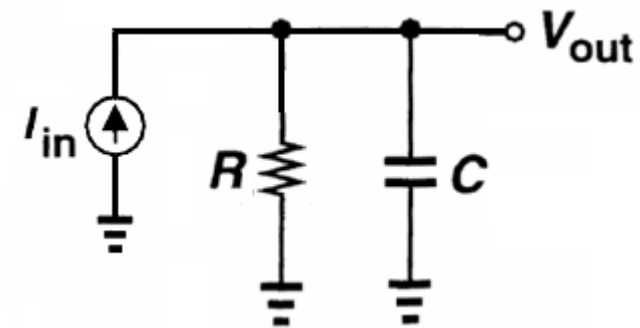
Using current input

$$H(s) = \frac{V_{out}}{I_{in}}(s) = \frac{1}{sC + 1/R} = \frac{R}{1 + sRC} = \frac{R}{1 + s / \omega_p}$$

$$\text{Let } D = 1 + sRC = 0 \Rightarrow s_p = -\frac{1}{RC} = -\omega_p$$

$$H(j\omega) = \frac{V_{out}}{I_{in}}(j\omega) = \frac{R}{1 + j\omega RC} = \frac{R}{1 + j\omega / \omega_p}$$

Difference is that gain at DC is R
Others are the same

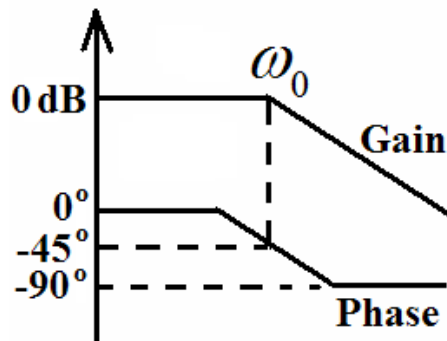


Poles and Zeros

LHP:

$$\frac{1}{1 + s / \omega_0}$$

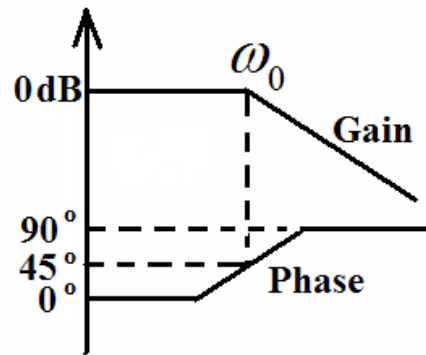
$$\frac{1}{1 + j(\omega / \omega_0)}$$



RHP:

$$\frac{1}{1 - s / \omega_0}$$

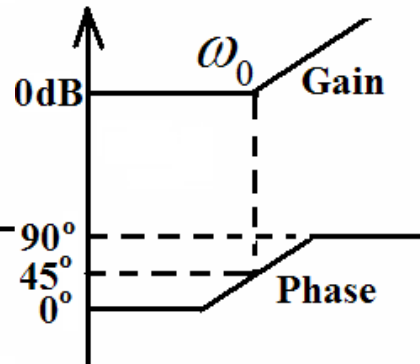
$$\frac{1}{1 - j(\omega / \omega_0)}$$



LHZ:

$$1 + s / \omega_0$$

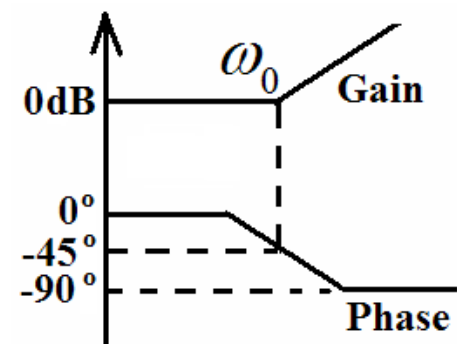
$$1 + j(\omega / \omega_0)$$



RHZ:

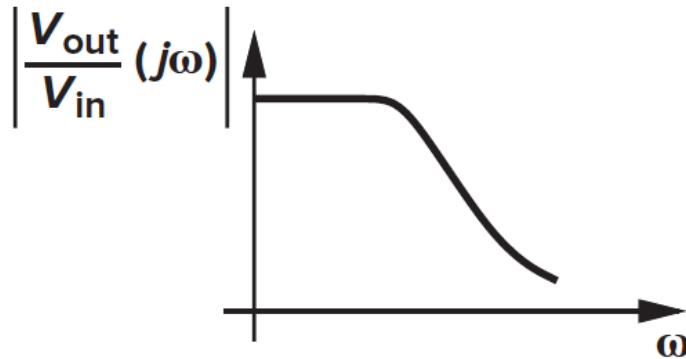
$$1 - s / \omega_0$$

$$1 - j(\omega / \omega_0)$$

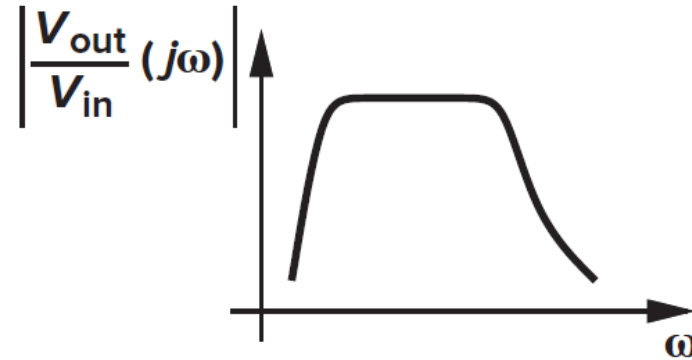


Magnitude of typical frequency responses

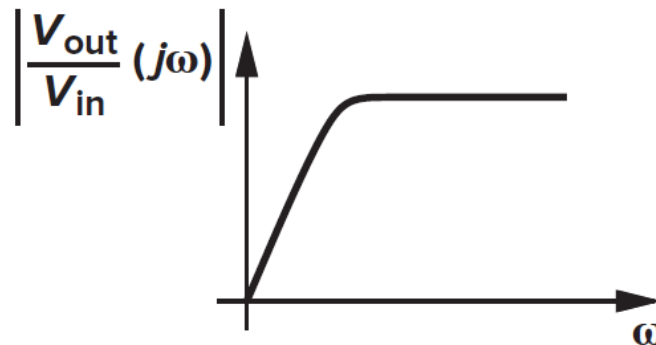
Low pass



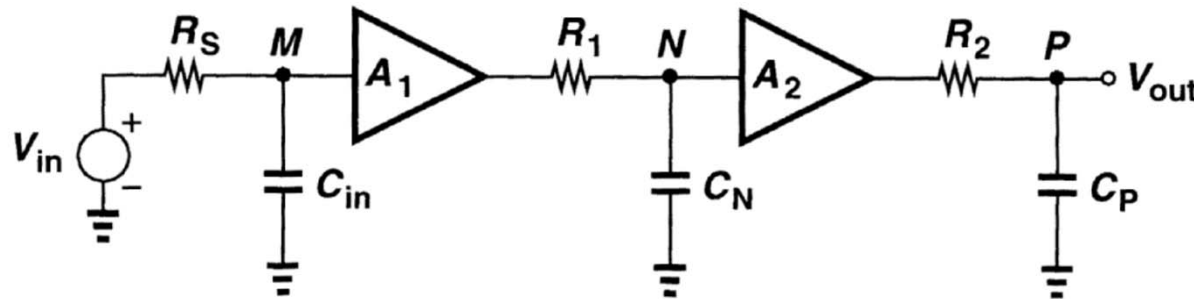
Band pass



High pass



Association of Poles and Nodes



$$\frac{V_{out}}{V_{in}}(s) = \frac{1}{1 + R_S C_{in} s} \cdot \frac{A_1}{1 + R_1 C_N s} \cdot \frac{A_2}{1 + R_2 C_P s}$$

$$= 1 \cdot A_1 A_2 \frac{1}{1 + s / \omega_{pM}} \cdot \frac{1}{1 + s / \omega_{pN}} \cdot \frac{1}{1 + s / \omega_{pP}}$$

Each node contribute one pole to the transfer function

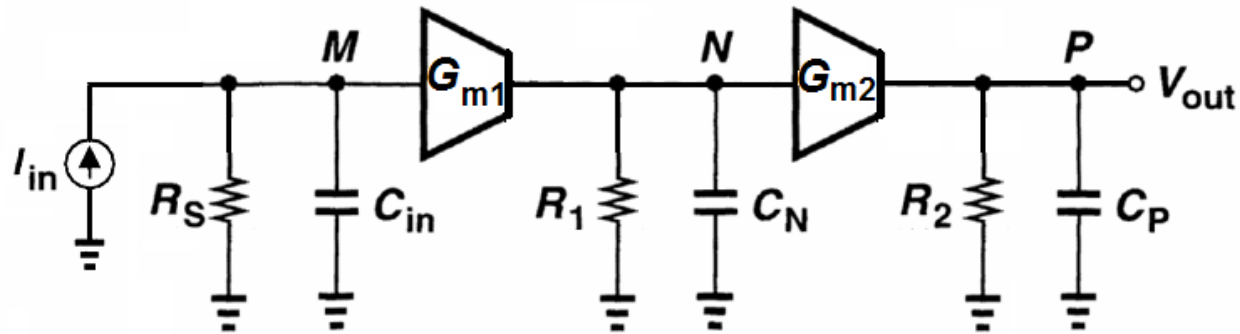
$$\omega_{p,j} = 1 / (R_j C_j)$$

R_j is the resistance to ground, C_j is the total cap.



Association of poles and nodes

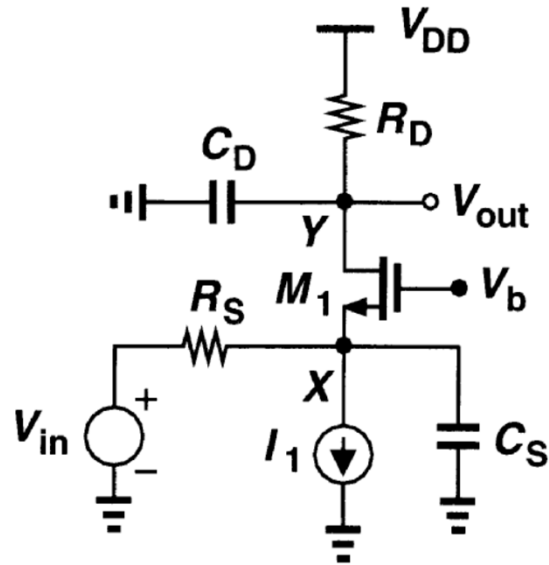
Parallel RC situation



$$\frac{V_{out}}{I_{in}}(s) = R_S \cdot G_{m1} R_1 \cdot G_{m2} R_2 \frac{1}{1 + s / \omega_{pM}} \cdot \frac{1}{1 + s / \omega_{pN}} \cdot \frac{1}{1 + s / \omega_{pP}}$$



Frequency response of Common Gate stage



If $\lambda = 0$,

$$\omega_{pX} = \frac{1}{(C_{GS} + C_{SB})(R_S \parallel \frac{1}{g_m + g_{mb}})}$$

$$\omega_{pY} = \frac{1}{(C_{GD} + C_{DB})R_D}$$

$$\frac{V_{out}}{V_{in}}(s) = \frac{(g_m + g_{mb})R_D}{1 + (g_m + g_{mb})R_S} \cdot \frac{1}{(1 + \frac{s}{\omega_{pX}})(1 + \frac{s}{\omega_{pY}})}$$



Outline

- General consideration
- Common-source stage
 - Miller-effect
 - Frequency response of CS stage
- Source Followers
- Common-Gate stage
- Cascode stage
- Differential Pair

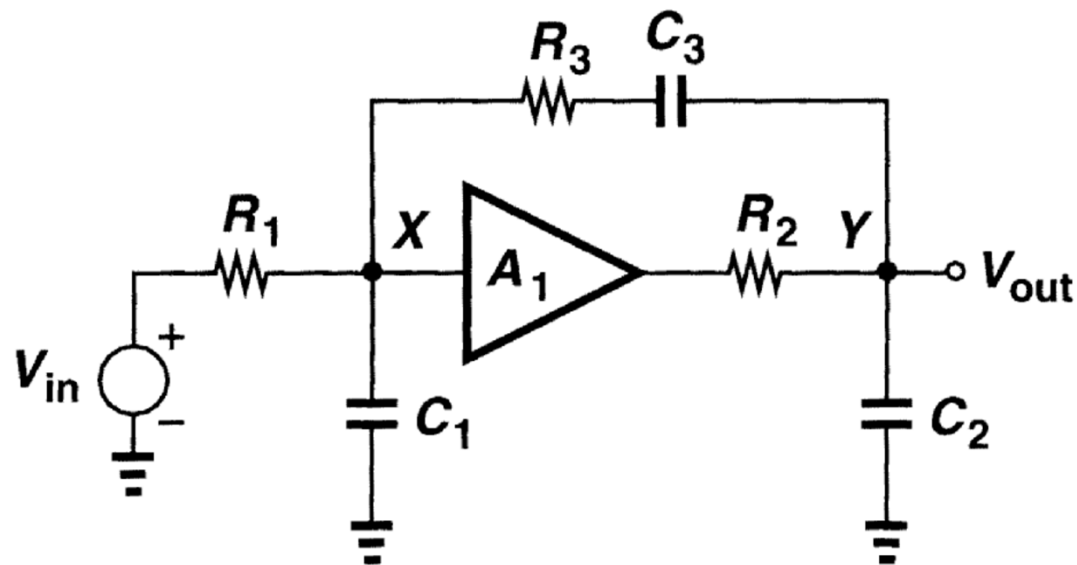


Poles and Nodes

Complicate situation

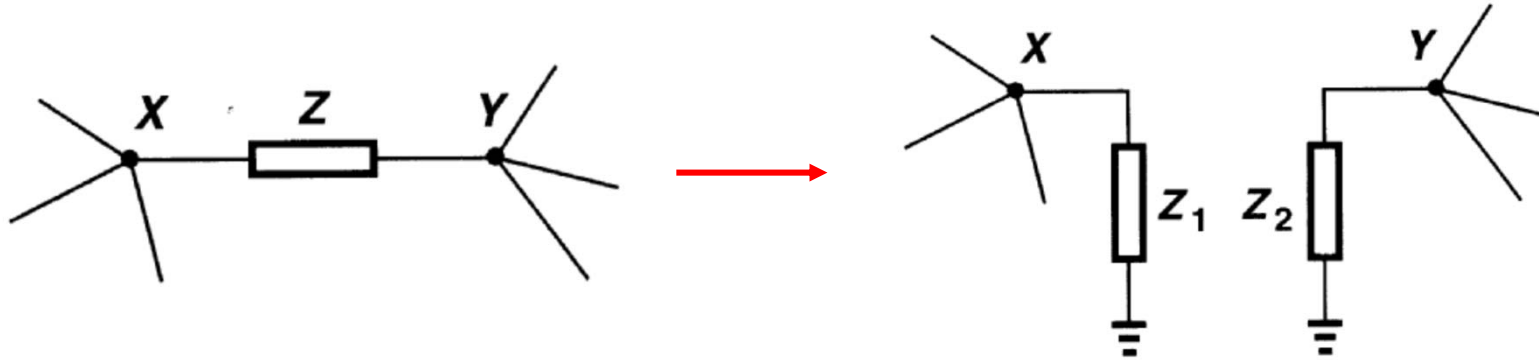
Feedback branch creates interaction between X and Y nodes

Difficult to determine R and C at the interaction nodes



Miller's Theorem

Used to decompose floating impedance



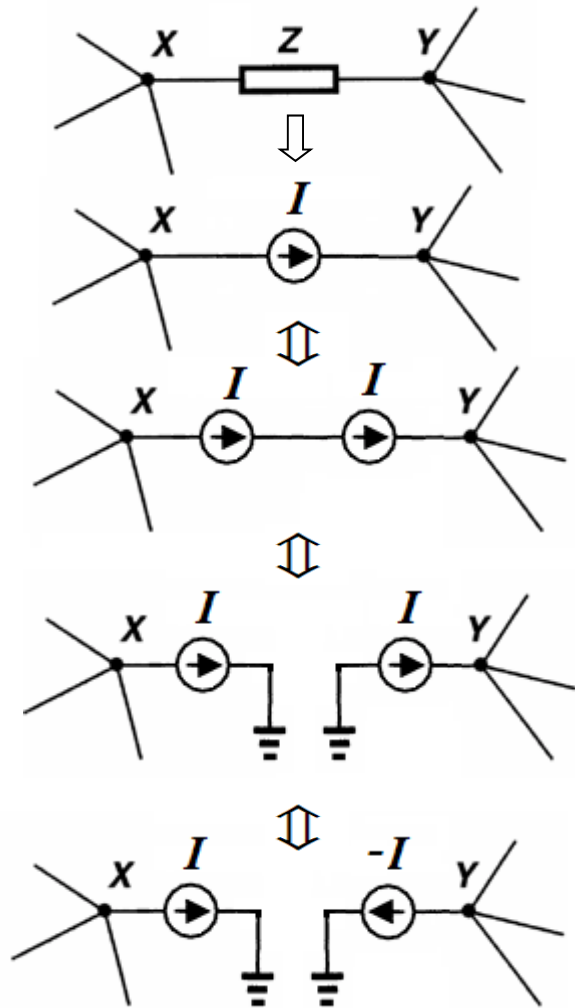
If the gain from X to Y is $A_v = \frac{V_Y}{V_X}$

Z can be decomposed into two impedances to ground

$$Z_1 = \frac{Z}{1 - A_v} \quad Z_2 = \frac{Z}{1 - A_v^{-1}}$$



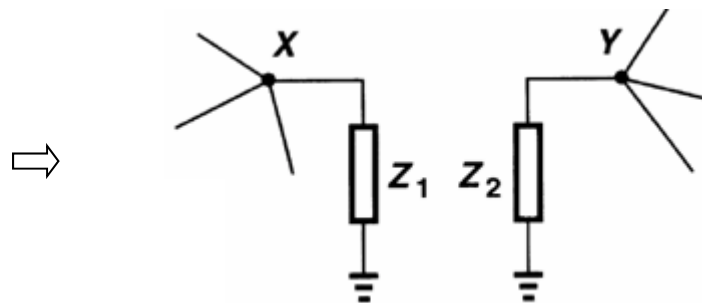
Proven of Miller Theorem



$$I = \frac{V_X - V_Y}{Z} = \frac{V_X (1 - A_v)}{Z} = \frac{V_X}{Z / (1 - A_v)}$$

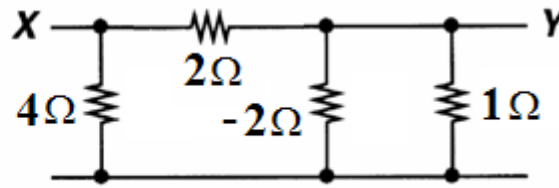
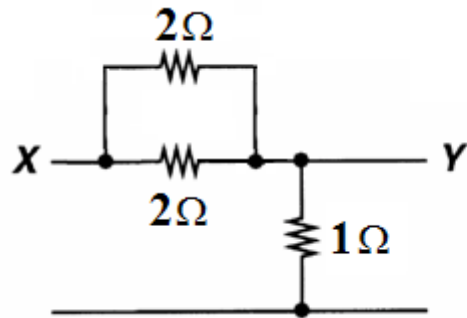
$$-I = \frac{V_Y - V_X}{Z} = \frac{V_Y (1 - A_v^{-1})}{Z} = \frac{V_Y}{Z / (1 - A_v^{-1})}$$

$$\Rightarrow Z_1 = \frac{Z}{1 - A_v}, Z_2 = \frac{Z}{1 - A_v^{-1}}$$

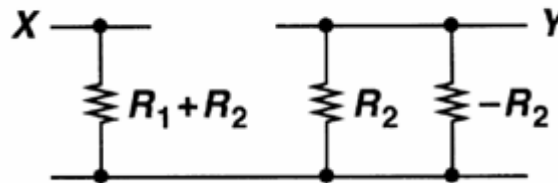
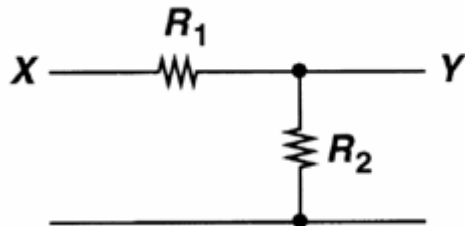


Applicability of Miller's Theorem

Decompose the upper resistor

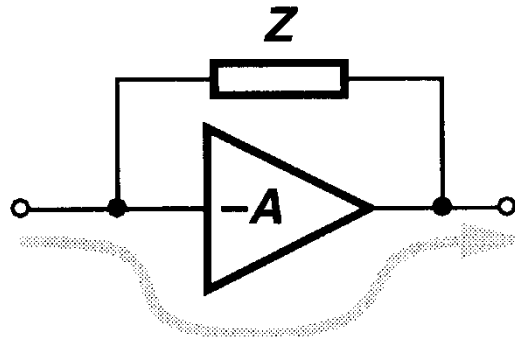


If Z forms the only path from X to Y, Miller Theorem is not applicable



Application of Miller's Theorem

Typical case: decompose the impedance parallel to the main signal path



Main Signal Path

Usually, both Z and A are functions of frequency.

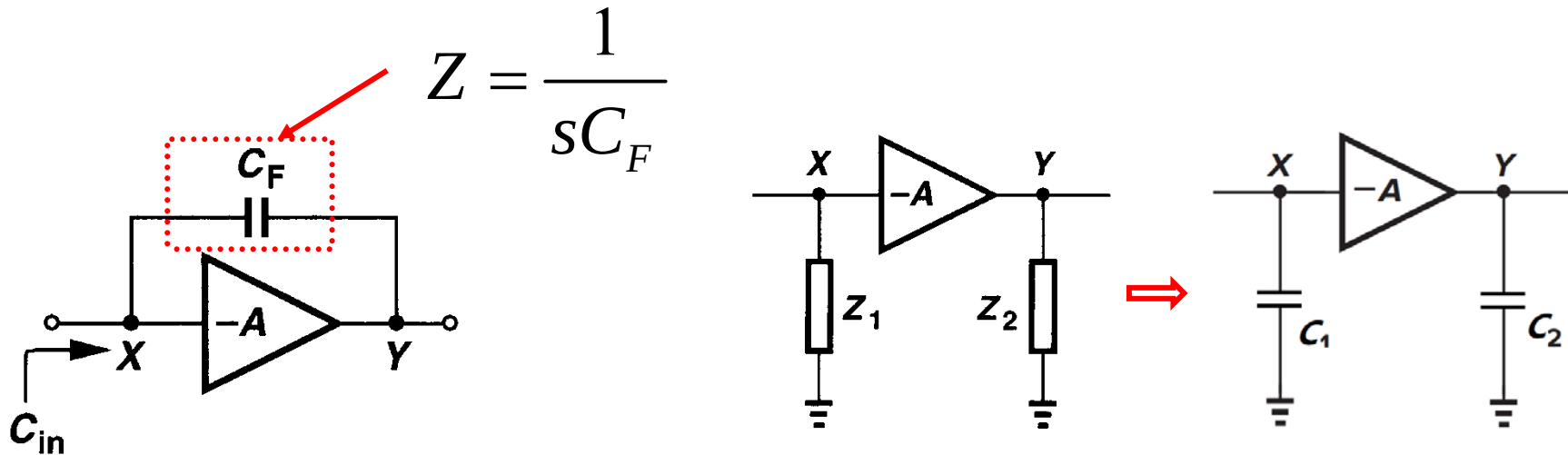
Accurate calculation is very complex

Often use the value of A at DC to decompose Z



General application: Ex. 6.1

- Calculating the input capacitance

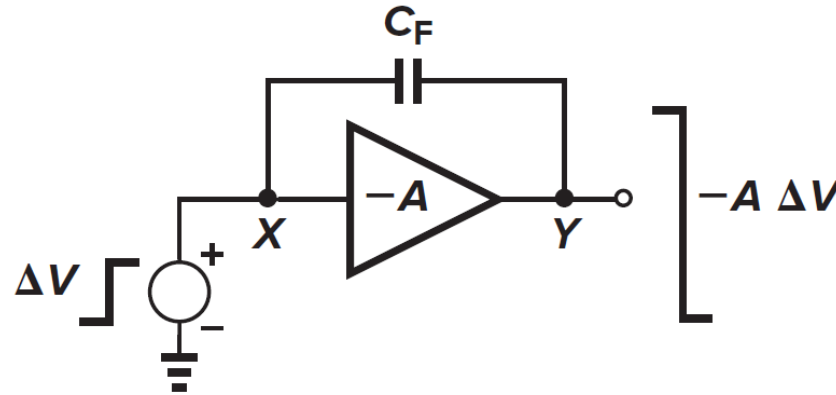


$$Z_1 = \frac{Z}{1 - (-A)} = \frac{1}{sC_F \cdot (1 + A)} \quad \longrightarrow \quad C_{in} = C_1 = (1 + A) \cdot C_F$$

$$Z_2 = \frac{Z}{[1 - (-A)^{-1}]} \approx \frac{1}{sC_F} \quad \longrightarrow \quad C_2 \approx C_F$$



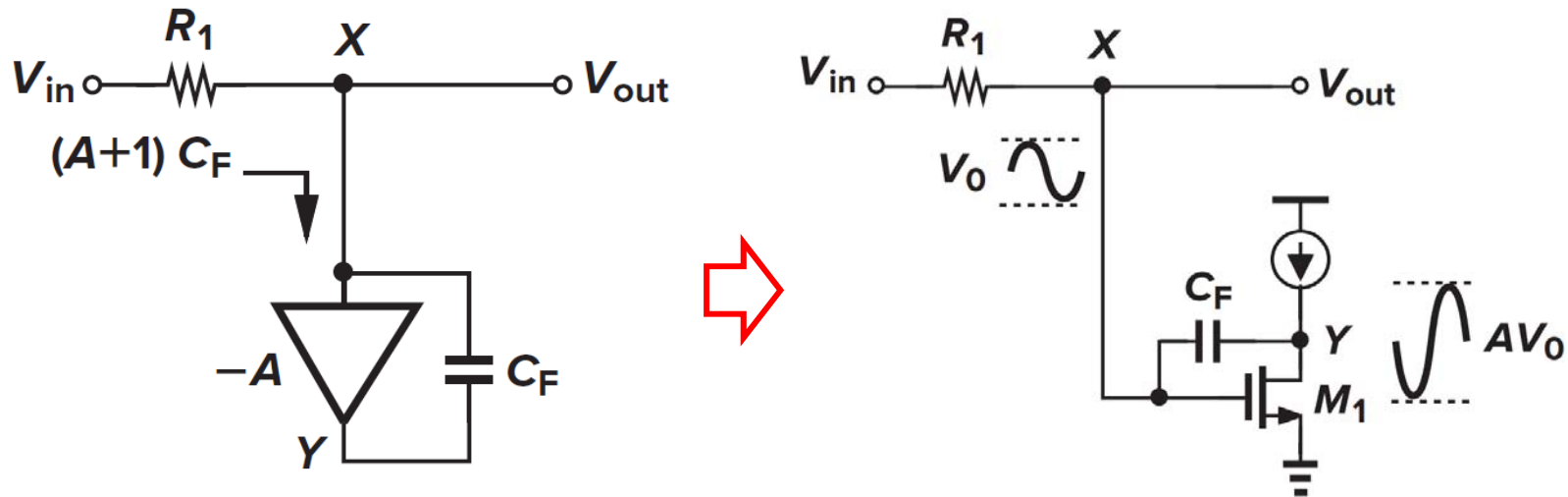
Why is C_F multiplied by $(1+A)$?



- To measure the input capacitance
 - Applying a **voltage step ΔV** , calculating the **charge** supplied by the voltage source
- Input is $\Delta V \rightarrow$ output is $-A \cdot \Delta V$
 - $\rightarrow$ total voltage change is $(1+A)\Delta V$
- The charge drawn from V_{in} is $(1+A)C_F \Delta V$
- Equivalent capacitance is $(1+A)C_F$



Realizing large capacitor using Miller effect



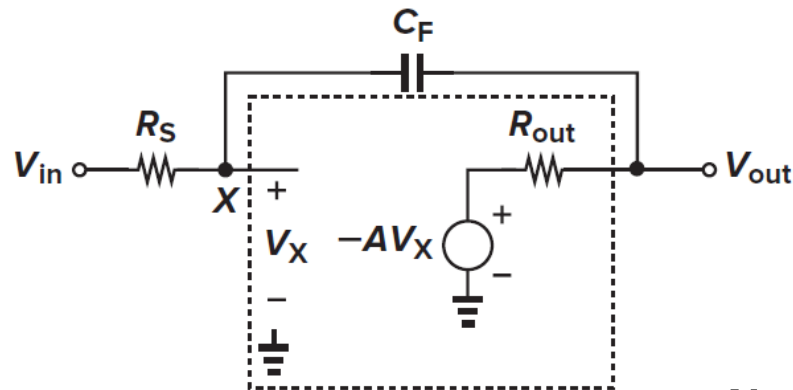
• Ex. 6.2

- RC filter with pole frequency decreased by $(1+A)$
- a useful method to implement large capacitors.
- **What are the drawbacks?**
 - Signal swing at Y is increased by A times, the amplifier must be able to handle this large swing.
 - Larger Power consumption.



Ex 6.4

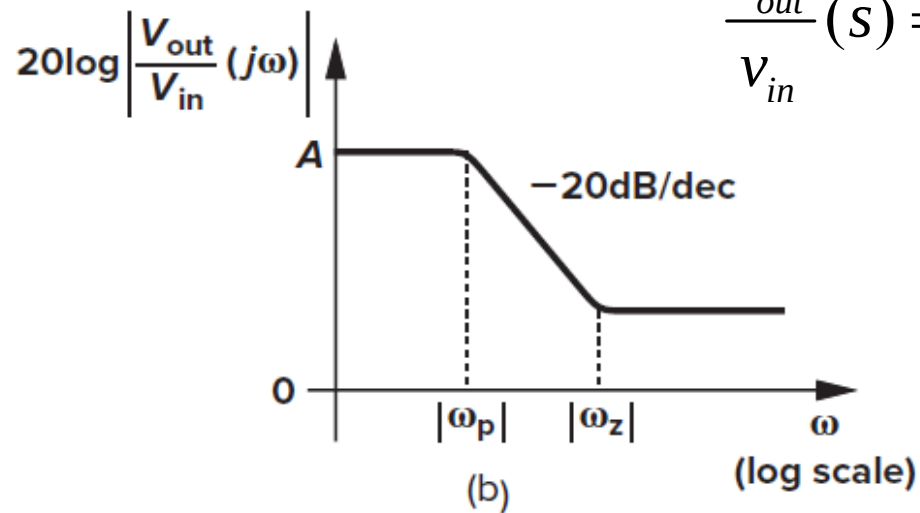
- Determine the transfer function using (a) direct analysis; (b) Miller approximation



$$\begin{cases} \frac{V_{in} - V_x}{R_S} R_{out} - A V_x = V_{out} \\ \frac{V_{in} - V_x}{R_S} = (V_x - V_{out}) \cdot s C_F \end{cases}$$



$$\frac{V_{out}}{V_{in}}(s) = A \frac{s(R_{out} C_F) / A - 1}{[(A+1)C_F R_S + R_{out} C_F] s + 1}$$



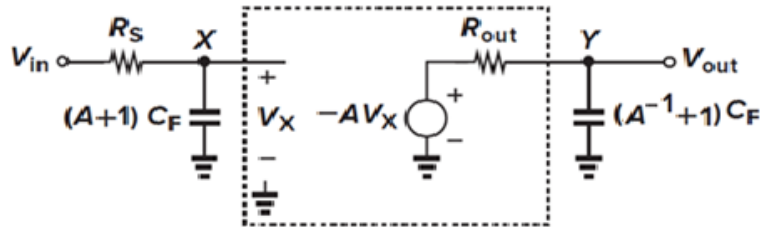
$$\omega_z = + \frac{1}{(R_{out} C_F) / A}$$

$$\omega_p = \frac{1}{(A+1)C_F R_S + R_{out} C_F}$$



Miller approximation

v.s Direct analysis



$$\frac{V_{out}}{V_{in}}(s) = -A \frac{s(R_{out}C_F)/A - 1}{[(A+1)C_F R_S + R_{out}C_F]s + 1}$$

$$\frac{V_X}{V_{in}}(s) = \frac{1}{(A+1)C_F R_S s + 1} \Rightarrow \frac{V_{out}}{V_{in}}(s) = \frac{-A}{[(A+1)C_F R_S s + 1][(A^{-1}+1)C_F R_{out} s + 1]}$$

- **Inaccuracy of Miller approximation**
 - The zero is lost and an additional pole appears
 - The dominant pole frequency is also not accurate
 - Why?
- Even though, Miller appr. is still a very useful method, especially for estimating the input capacitance



Outline

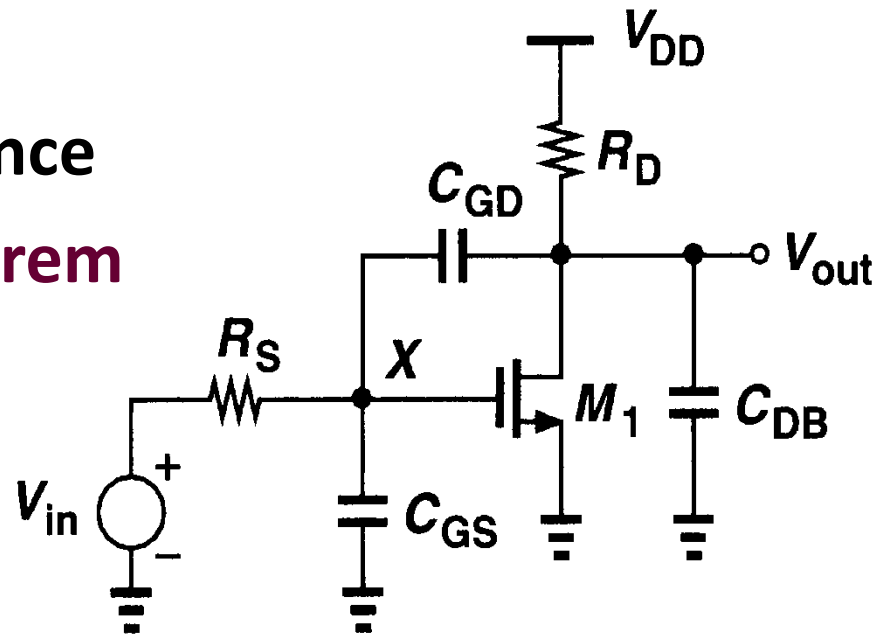
- General consideration
- Common-source stage
 - Miller-effect
 - **Frequency response of CS stage**
- Source Followers
- Common-Gate stage
- Cascode stage
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Frequency response of common-source stage

- Analyze the effect of parasitic capacitors
 - Derive the transfer function of $A_v(s)$
 - Finding out the poles
- Considering all the parasitic capacitances
 - Grounded: C_{GS} , C_{DB}
 - C_{DS} is a floating impedance
 - Applying Miller Theorem

$$A_{v,DC} = -g_m R_D$$



Finding out poles(1)

- Case #1: R_S is relatively small
 - Because of Miller effect
 - Total capacitance at X node is $C_{GS} + (1 + g_m R_D) C_{GD}$
 - Small signal resistance at X node is R_S

$$\longrightarrow \omega_{p,in} = \frac{1}{R_S [C_{GS} + (1 + g_m R_D) C_{GD}]}$$

- Total cap. at the output is around $C_{GD} + C_{DB}$
- Resistance at the output node is about R_D

$$\longrightarrow \omega_{p,out} = \frac{1}{(C_{GD} + C_{DB}) R_D}$$

Finding out poles (2)

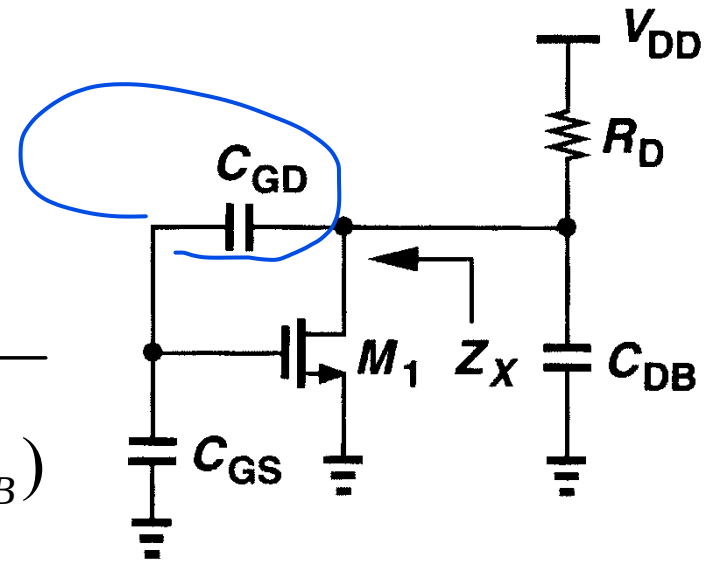
- **Case #2: R_S is relatively large**
 - The expression for the input pole ω_{in} is the same
 - However, the output resistance is changed

$$R_X = \frac{C_{GD} + C_{GS}}{C_{GD}} \frac{1}{g_m} \quad \text{total resistance is } R_X \parallel R_D \parallel r_o$$

- Define $C_{eq} = C_{GD}C_{GS}/(C_{GD} + C_{GS})$

- Total cap. is $C_{eq} + C_{DB}$

$$\omega_{p,out} = \frac{1}{\left[R_D \parallel \left(\frac{C_{GD} + C_{GS}}{C_{GD}} \cdot \frac{1}{g_{m1}} \right) \right] (C_{eq} + C_{DB})}$$



Transfer function

$$\omega_{p,in} = \frac{1}{R_S [C_{GS} + (1 + g_m R_D) C_{GD}]}$$

$$\omega_{p,out} = \frac{1}{(C_{GD} + C_{DB}) R_D}$$

R_S small

$$\omega_{p,out} = \frac{1}{\left[R_D \parallel \left(\frac{C_{GD} + C_{GS}}{C_{GD}} \cdot \frac{1}{g_{m1}} \right) \right] (C_{eq} + C_{DB})} \quad R_S \text{ large}$$

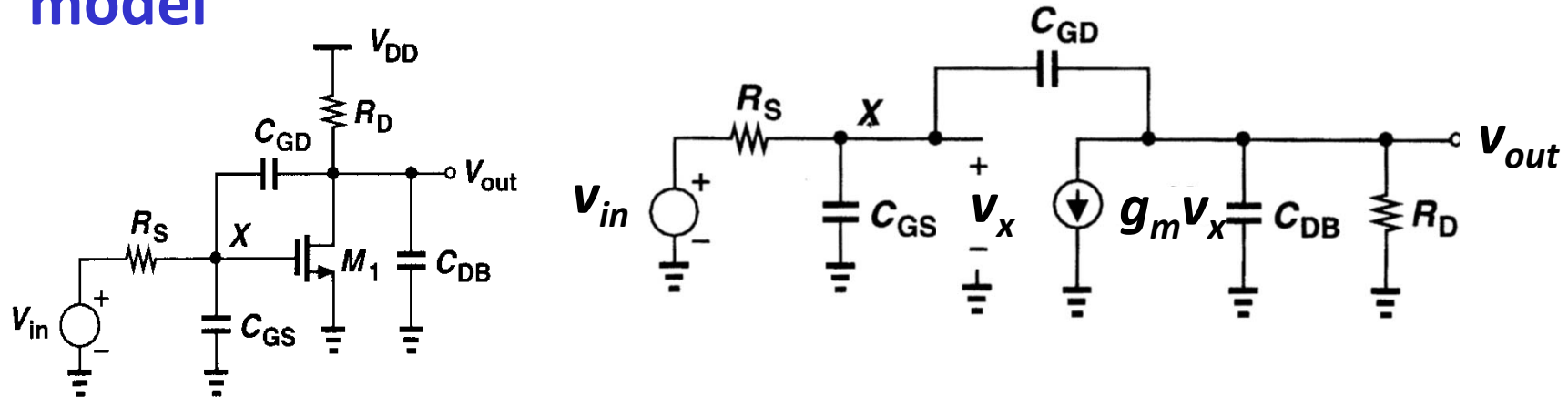
$$A_v(s) = \frac{V_{out}}{V_{in}}(s) = -g_m R_D \frac{1}{(1 + s / \omega_{in})(1 + s / \omega_{out})}$$

- Errors of miller analysis: (1) a zero is neglected; (2) gain is DC gain, not accurate. **Verified by accurate analysis**



Accurate analysis using small-signal model

- Obtain exact transfer function through small-signal model



$$\frac{v_x - v_{in}}{R_S} + v_x \cdot sC_{GS} + (v_x - v_{out}) \cdot sC_{GD} = 0$$

$$(v_{out} - v_x) \cdot sC_{GD} + g_m v_x + v_{out} \left(\frac{1}{R_D} + sC_{DB} \right) = 0$$

$v_x = \dots$

Exact transfer function

$$\longrightarrow v_x = -\frac{v_{out}(sC_{GD} + 1/R_D + sC_{DB})}{g_m - sC_{GD}}$$

$$\longrightarrow \frac{v_{out}}{v_{in}} = \frac{(sC_{GD} - g_m)R_D}{s^2 \cdot R_S R_D \cdot \zeta + s[R_S(1 + g_m R_D)C_{GD} + R_S C_{GS} + R_D(C_{GD} + C_{DB})] + 1}$$

where $\zeta = C_{GS}C_{GD} + C_{GS}C_{DB} + C_{GD}C_{DB}$

- The transfer function is 2nd-order even though there are 3 capacitors
 - They form a “loop”, only two independent nodes.

Simplification of the transfer function

- If the distance between the two poles is large

- $|\omega_{p1}| \ll |\omega_{p2}|$,

- The denominator is simplified to

$$D = \left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right) \approx \frac{s^2}{\omega_{p1}\omega_{p2}} + \frac{s}{\omega_{p1}} + 1$$

ω_{p1} can be obtained directly:

$$\omega_{p1} = \frac{1}{R_S [C_{GS} + (1 + g_m R_D) C_{GD}] + R_D (C_{GD} + C_{DB})}$$



Calculation of ω_{p2}

$$\omega_{p2} = \frac{1}{\omega_{p1}} \cdot \frac{1}{R_S R_D (C_{GS} C_{GD} + C_{GS} C_{DB} + C_{GD} C_{DB})}$$

$$= \frac{R_S (1 + g_m R_D) C_{GD} + R_S C_{GS} + R_D (C_{GD} + C_{DB})}{R_S R_D (C_{GS} C_{GD} + C_{GS} C_{DB} + C_{GD} C_{DB})}$$

If C_{GS} is large enough

$$\omega_{p2} = \frac{\cancel{R_S (1 + g_m R_D) C_{GD}} + R_S C_{GS} + \cancel{R_D (C_{GD} + C_{DB})}}{R_S R_D (C_{GS} C_{GD} + C_{GS} C_{DB} + \cancel{C_{GD} C_{DB}})}$$

$$\rightarrow \omega_{p2} \approx \frac{1}{R_D (C_{GD} + C_{DB})}$$

The same as miller equivalent when R_S is small

If C_{GD} is large enough

$$\omega_{p2} = \frac{R_S(1 + g_m R_D)C_{GD} + \cancel{R_S C_{GS}} + \cancel{R_D(C_{GD} + C_{DB})}}{R_S R_D (C_{GS} C_{GD} + \cancel{C_{GS} C_{DB}} + C_{GD} C_{DB})}$$

$$\longrightarrow \omega_{p2} \approx \frac{g_m}{C_{GS} + C_{DB}}$$

In reality, C_{GS} is always larger than C_{GD} .

However, in most applications, a capacitor parallel to C_{GD} is used to obtain a large enough ω_{p2} .

The zero in the transfer function

$$\frac{V_{out}}{V_{in}} = \frac{(sC_{GD} - g_m)R_D}{s^2 \cdot R_S R_D \cdot \zeta + s[R_S(1 + g_m R_D)C_{GD} + R_S C_{GS} + R_D(C_{GD} + C_{DB})] + 1}$$

Let the numerator = 0

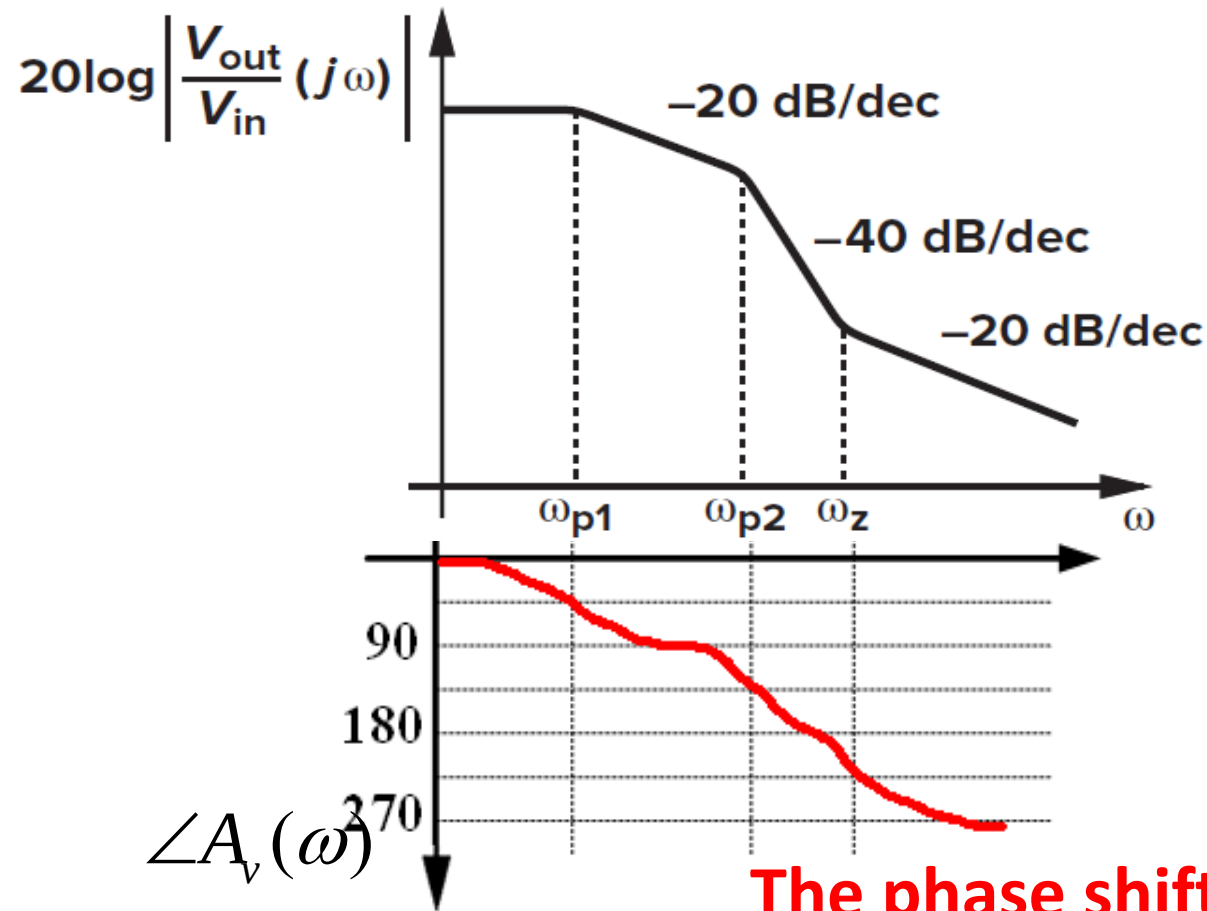
$$sC_{GD} - g_m = 0 \rightarrow s_z = \frac{g_m}{C_{GD}} = +\omega_z$$

A right half plane zero, harmful to the stability !!
(See chapter 10)



Effects of the RHP zero

The zero cancels the effect of the pole

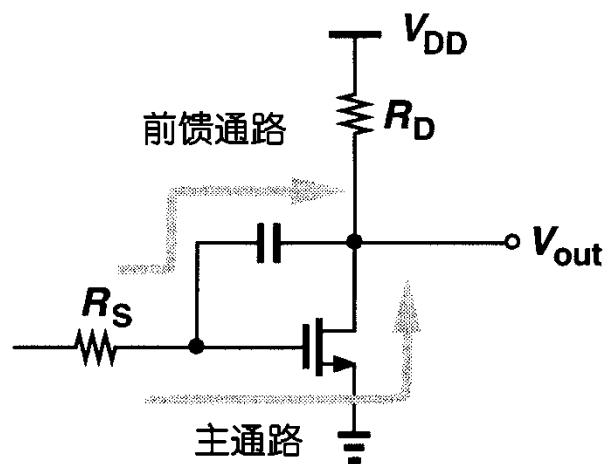


The phase shift is increased!!

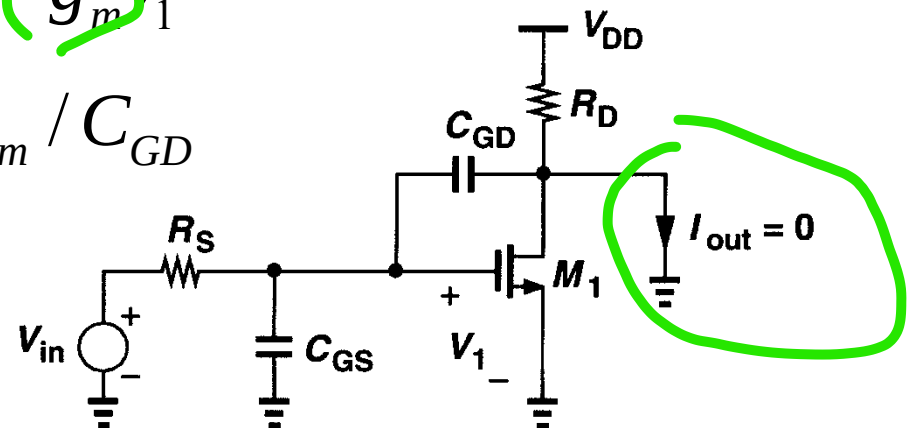


Where is the zero from?

- C_{GD} provides a feedforward path
 - The input is directly coupled to the output
- A simple calculation method
 - At the **complex frequency** s_z , $v_{out}=0$
 - Output is shorted, no current flows to output



$$v_1 C_{GD} s_z = g_m v_1$$
$$\rightarrow s_z = g_m / C_{GD}$$

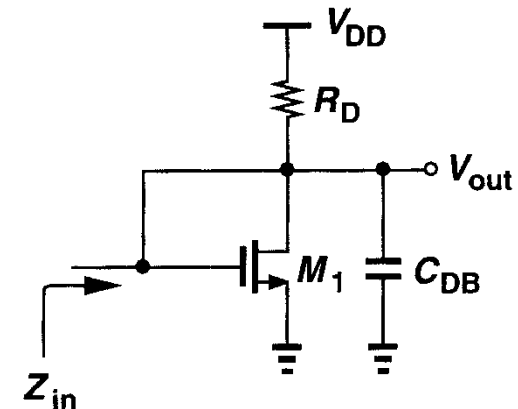
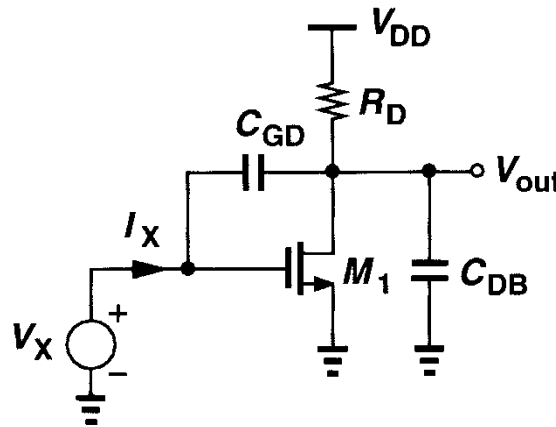
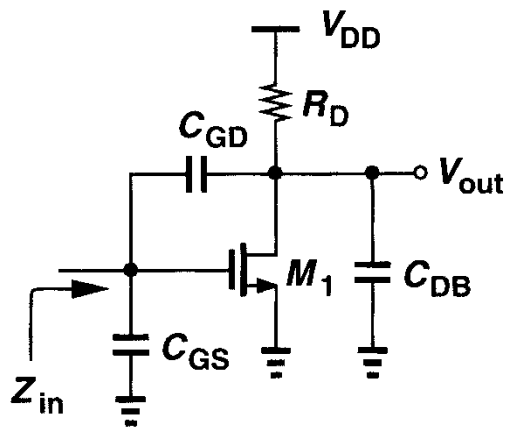


CS input impedance

- Because of Miller

$$Z_{in} = \frac{1}{s \cdot [C_{GS} + (1 + g_m R_D) C_{GD}]}$$

- At high frequencies, feedforward influences Z_{in}

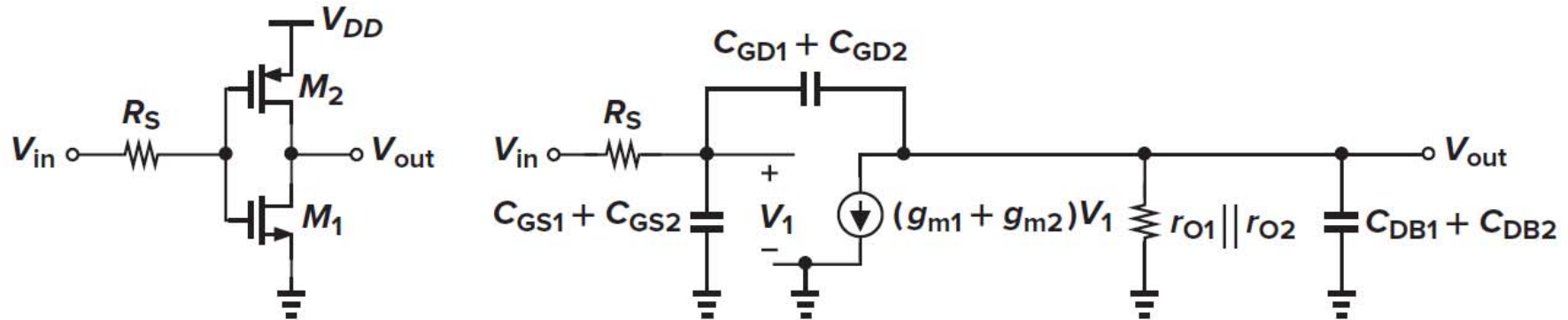


$$Z_{in} = \frac{V_X}{i_X} = \frac{1 + R_D (C_{GD} + C_{DB}) s}{s C_{GD} (1 + g_m R_D + s R_D C_{DB})} \parallel \frac{1}{s C_{GS}}$$

Pure capacitive
to complex
impedance !



Class-AB stage



- The same structure as the C_s stage, but with
 - Larger C_{GS} , C_{GD}
 - Higher g_m



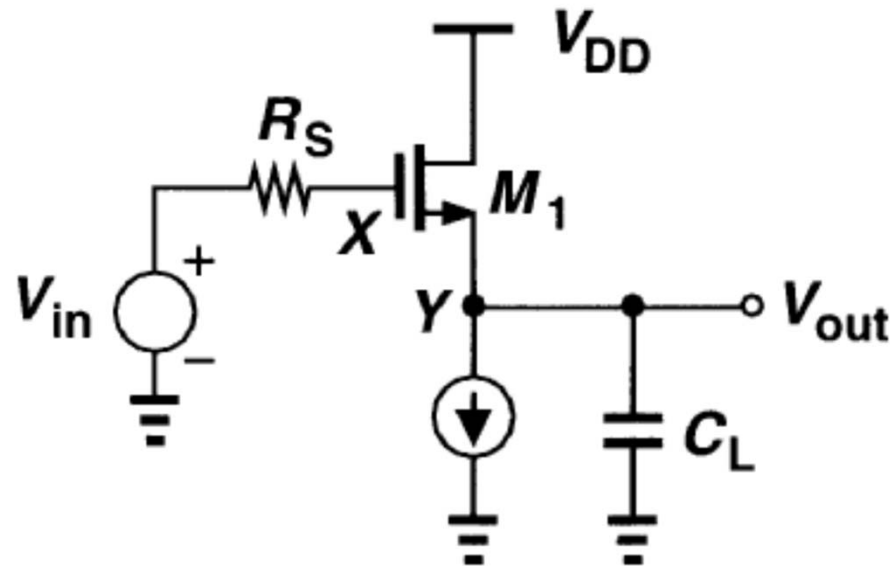
Outline

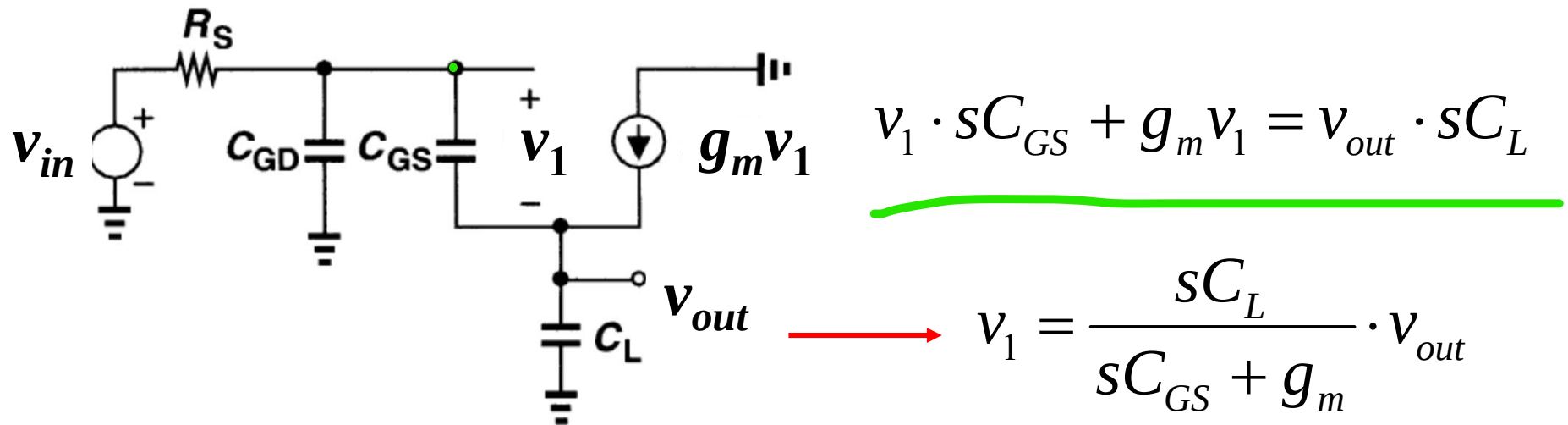
- General consideration
- Common-source stage
- **Source Followers**
- Common-Gate stage
- Cascode stage
- Differential Pair



Source Followers (Common Drain)

- Source followers are used as voltage buffers, their frequency response influences the overall response
- Because of C_{GS} , difficult to associate a pole with each nodes, using small-signal model





$$v_{in} = R_S [v_1 \cdot sC_{GS} + (v_1 + v_{out}) \cdot sC_{GD}] + v_1 + v_{out}$$

$$\frac{v_{out}}{v_{in}} = \frac{g_m + sC_{GS}}{s^2 R_S \cdot \zeta + s(g_m R_S C_{GD} + C_L + C_{GS}) + g_m}$$

$$\zeta = C_L C_{GD} + C_{GD} C_{GS} + C_{GS} C_L$$



ω_z and ω_{p1}

$$\frac{V_{out}}{V_{in}} = \frac{g_m + sC_{GS}}{s^2 R_S \cdot \zeta + s(g_m R_S C_{GD} + C_L + C_{GS}) + g_m}$$

$$\longrightarrow s_z = -\frac{g_m}{C_{GS}} = -\omega_z \quad \text{A LHP zero!}$$

- If the distance between the two poles is large

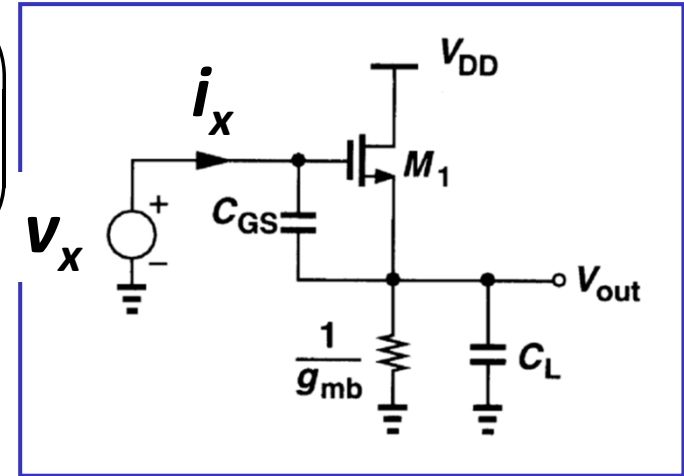
– If $|\omega_{p1}| \ll |\omega_{p2}|$,

$$\longrightarrow \omega_{p1} = \frac{g_m}{g_m R_S C_{GD} + C_L + C_{GS}}$$

Source Follower Input Impedance

$$v_x = \frac{i_x}{sC_{GS}} + \left(i_x + \frac{i_x}{sC_{GS}} \cdot g_m \right) \left(\frac{1}{g_{mb}} \parallel \frac{1}{sC_L} \right)$$

$$\rightarrow Z_{in} = \frac{1}{sC_{GS}} + \left(1 + \frac{g_m}{sC_{GS}} \right) \frac{1}{g_{mb} + sC_L}$$



- At low frequencies, $g_{mb} \gg |sC_L|$

$$Z_{in} \approx \frac{1}{sC_{GS}} \left(1 + g_m / g_{mb} \right) + 1 / g_{mb}$$

$$\rightarrow C_{in} = C_{GS} g_{mb} / (g_m + g_{mb}) \text{ (same as Miller)}$$

- C_{GS} is scaled by a factor

Input Impedance at high frequencies

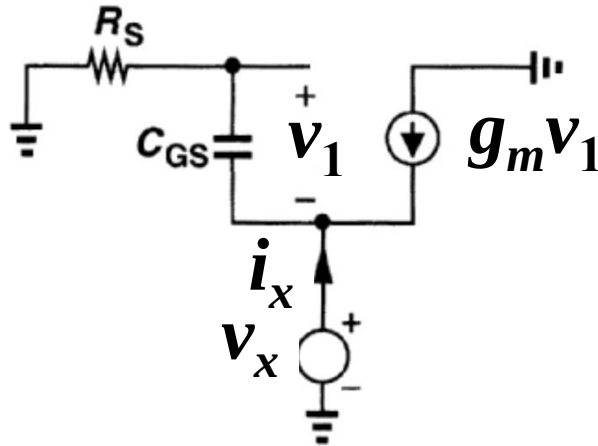
$$Z_{in} = \frac{1}{sC_{GS}} + \left(1 + \frac{g_m}{sC_{GS}}\right) \frac{1}{g_{mb} + sC_L}$$

At high frequencies, $g_{mb} \ll |sC_L|$

$$Z_{in} \approx \frac{1}{sC_{GS}} + \frac{1}{sC_L} + \frac{g_m}{s^2 C_{GS} C_L}$$

At a particular frequency, input impedance includes C_{GD} in parallel with series combination of C_{GS} and C_L and a negative resistance equal to $-g_m/(C_{GS}C_L\omega^2)$.

Source Follower Output Impedance

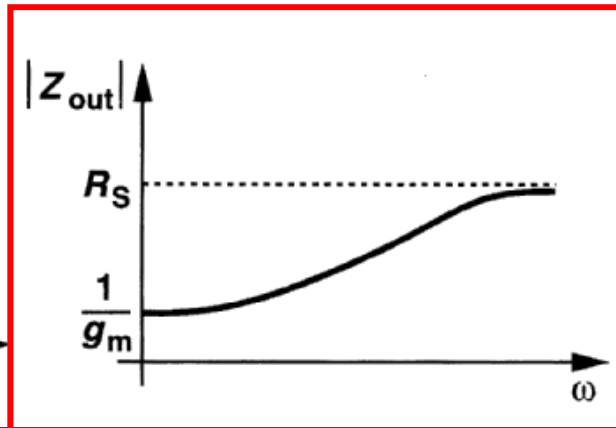
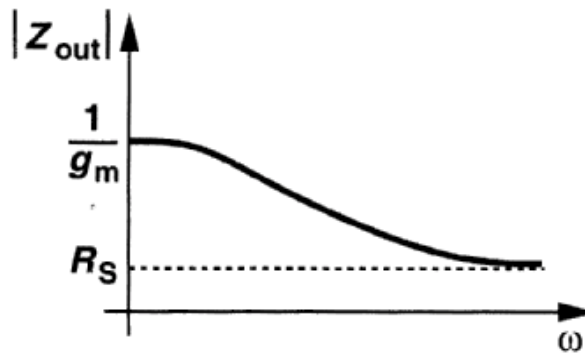


$$Z_{out} = v_x / i_x$$

$$= \frac{sR_S C_{GS} + 1}{g_m + sC_{GS}}$$

$$\approx 1/g_m, \text{ at low frequencies}$$

$$\approx R_S, \text{ at high frequencies}$$

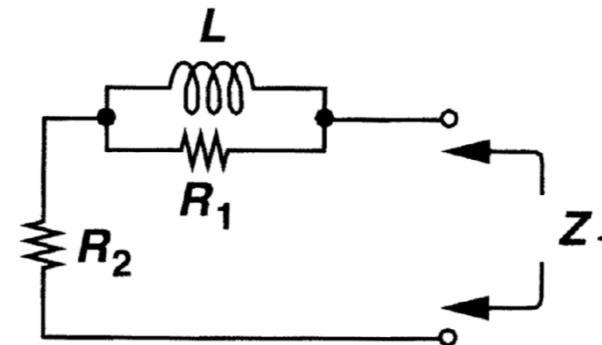
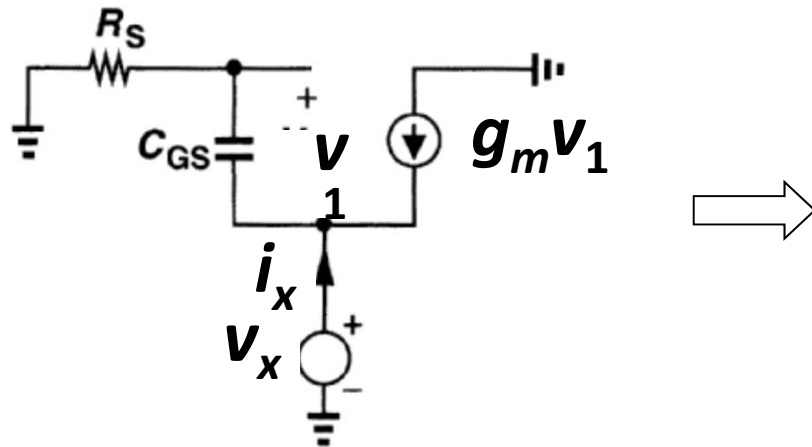


For CMOS,
 R_S always $> 1/g_m$



Source Follower Output Impedance (cont.)

Equivalent circuit



$$R_2 = 1/g_m$$

$$R_1 = R_S - 1/g_m$$

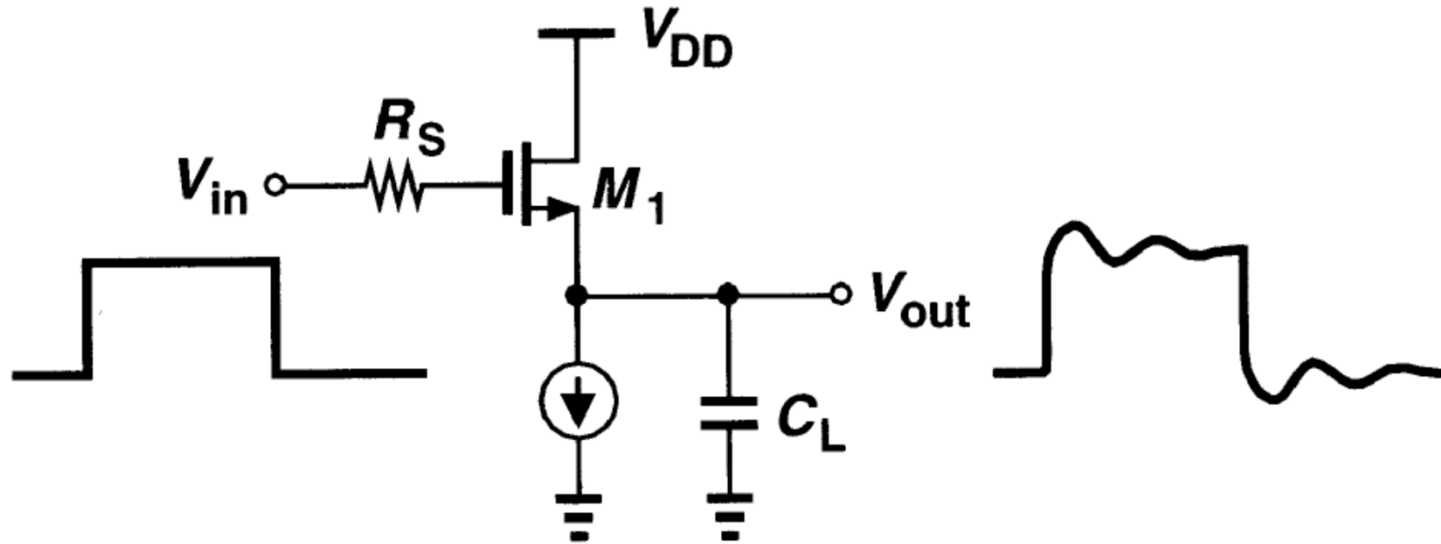
$$L = \frac{C_{GS}}{g_m} (R_S - 1/g_m)$$

$$Z_{out} = \frac{sR_S C_{GS} + 1}{g_m + sC_{GS}} \approx \begin{cases} 1/g_m, & \text{for LF} \\ R_S, & \text{for HF} \end{cases}$$

- Output inductance is dependent on source impedance, R_S !
- We can use the source follower to realize active inductor, see Ex. 6.14



Source Follower Ringing



Output ringing due to tuned circuit formed with C_L and inductive component of output impedance.

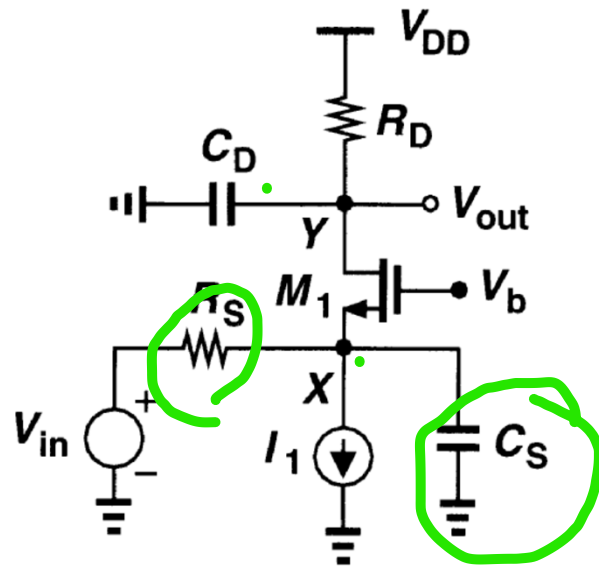


Outline

- General consideration
- Common-source stage
- Source Followers
- **Common-Gate stage**
- Cascode stage
- Differential Pair



Common Gate stage



If $\lambda = 0$,

$$\omega_{pX} = \frac{1}{(C_{GS} + C_{SB})(R_S \parallel \frac{1}{g_m + g_{mb}})}$$

$$\omega_{pY} = \frac{1}{(C_{GD} + C_{DB})R_D}$$

$$\frac{V_{out}}{V_{in}}(s) = \frac{(g_m + g_{mb})R_D}{1 + (g_m + g_{mb})R_S} \cdot \frac{1}{(1 + \frac{s}{\omega_{pX}})(1 + \frac{s}{\omega_{pY}})}$$



Outline

- General consideration
- Common-source stage
- Source Followers
- Common-Gate stage
- **Cascode stage**
- Differential Pair



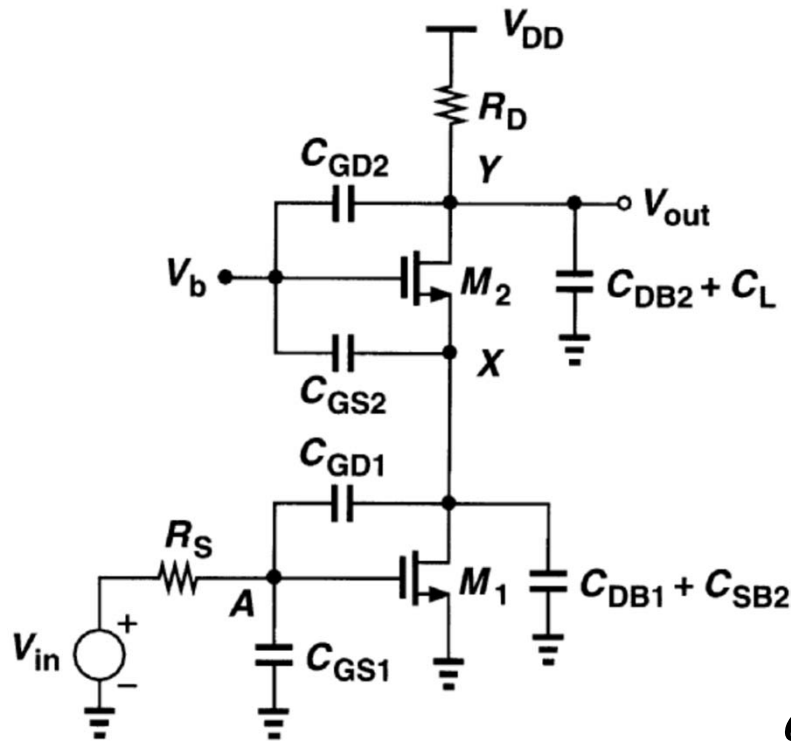
Cascode Stage

Three poles corresponding to the three nodes:
A, X, and Y

Resistance at A: R_S

Capacitance at A:

$C_{GS1} + (\text{miller equivalent of } C_{GD1})$



$$A_{v,A \text{ to } X} = -g_{m1} \cdot \frac{1}{g_{m2} + g_{mb2}}$$

$$\omega_{p,A} = \frac{1}{R_S \left[C_{GS1} + C_{GD1} \left(1 + \frac{g_{m1}}{g_{m2} + g_{mb2}} \right) \right]}$$



$$\omega_{p,X} \text{ and } \omega_{p,Y}$$

Resistance at X: $1/(g_{m2}+g_{mb2})$

Capacitance at X: C_{GS2} , C_{DB1} , C_{SB2} plus the miller equivalent of C_{GD1}

$$\omega_{p,X} \approx \frac{g_{m2} + g_{mb2}}{2C_{GD1} + C_{DB1} + C_{SB2} + C_{GS2}}$$

Resistance at Y: R_D

Capacitance at Y: C_{DB2} , C_{GD2} , C_L

$$\omega_{p,Y} = \frac{1}{R_D (C_{DB2} + C_L + C_{GD2})}$$

Transfer function of Cascode

$$A_v(s) = \frac{V_{out}}{V_{in}}(s) = \frac{-g_{m1}R_D}{(1 + s/\omega_{p,A})(1 + s/\omega_{p,Y})(1 + s/\omega_{p,X})}$$

- Usually, $\omega_{p,X}$ is much larger than $\omega_{p,A}$ and $\omega_{p,Y}$
- There is no zero in cascode
 - No feedthrough path exists
 - An important merit of cascode over CS stage



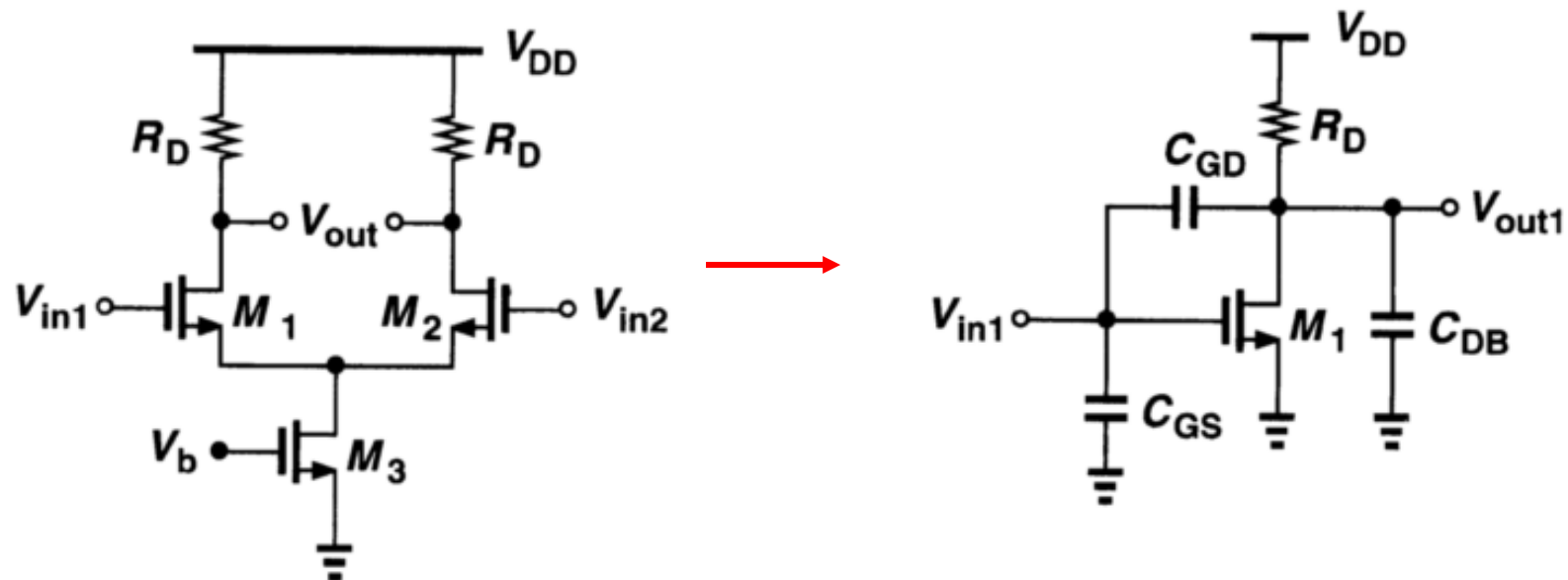
Outline

- General consideration
- Common-source stage
- Source Followers
- Common-Gate stage
- Cascode stage
- **Differential Pair**
 - **Differential pair with resistive load**
 - Current source load
 - Active current mirror



Resistive load: DM frequency response

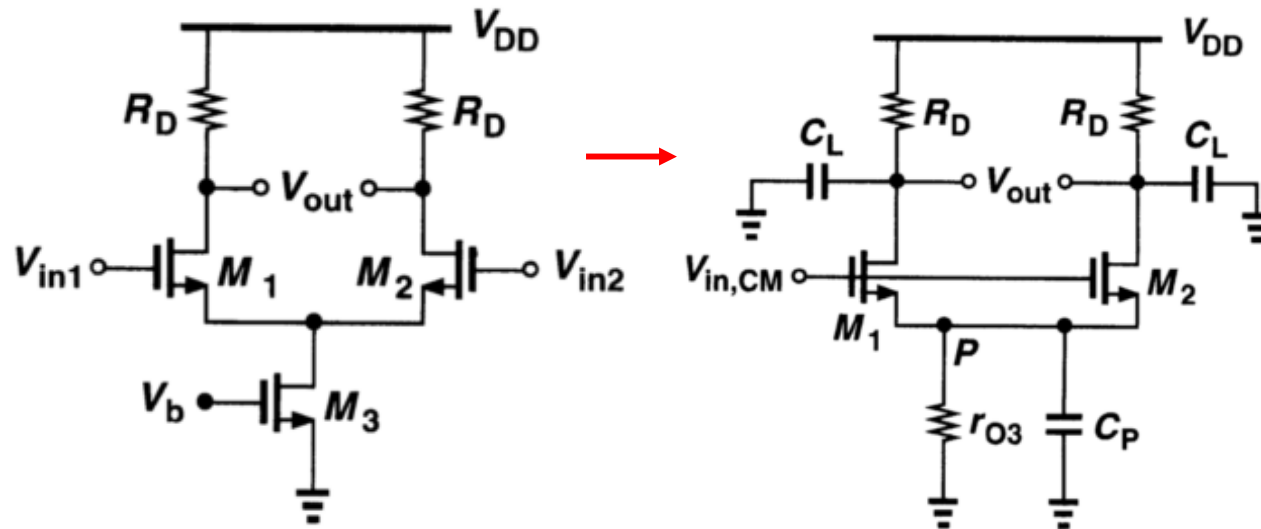
- Using half circuit equivalent



– Identical to a CS stage!!



Resistive load: CM gain frequency response

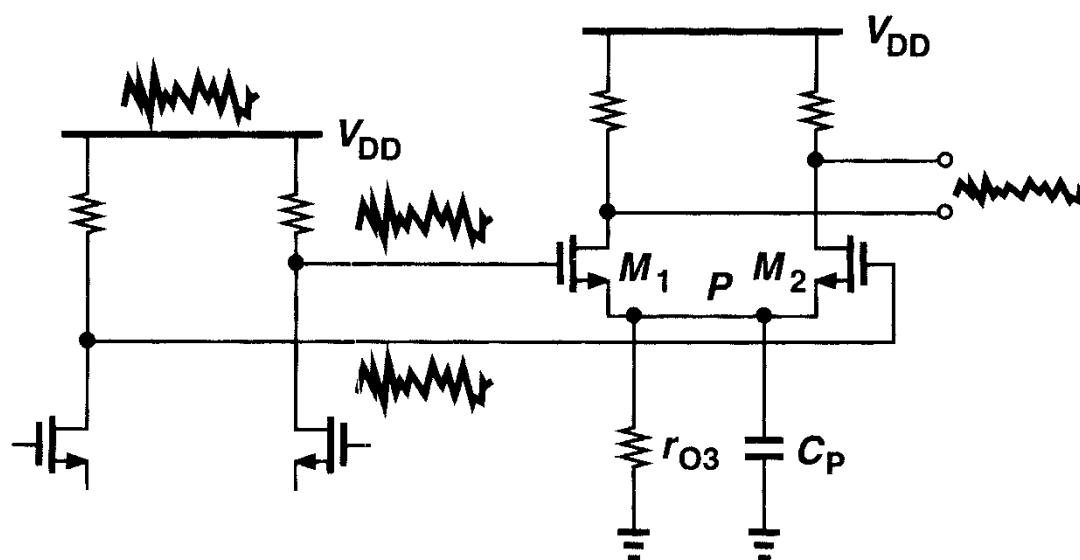


– Low frequency (Chapter 3)
$$A_{CM-DM} = \frac{\Delta g_m R_D}{1 + (g_{m1} + g_{m2})r_{o3}}$$

– High frequency
$$A_{CM-DM} = \frac{\Delta g_m \cdot (R_D \parallel 1/sC_L)}{1 + (g_{m1} + g_{m2}) \cdot (r_{o3} \parallel 1/sC_P)}$$

CMRR degrades at high frequencies

- C_p is large usually because M_3 has large size
 - CMRR degrades at high frequencies



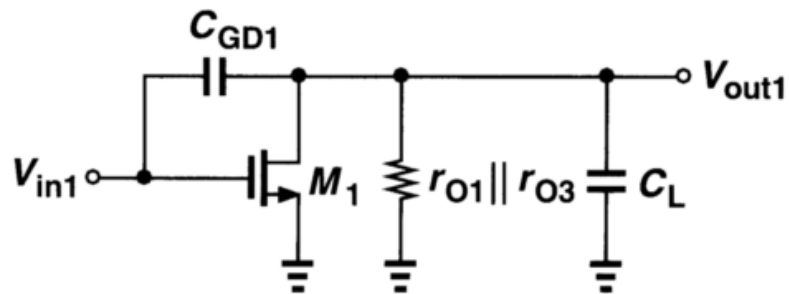
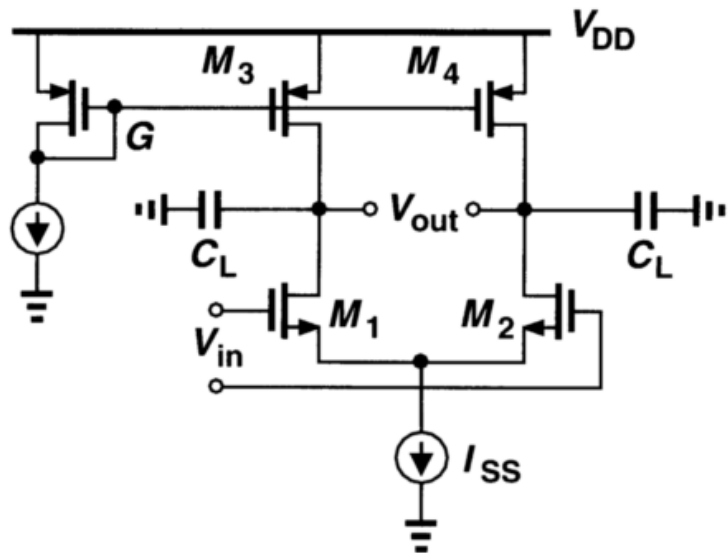
- Trade-off between signal range and CMRR
 - Choose the Size of M_3 properly

Outline

- General consideration
- Common-source stage
- Source Followers
- Common-Gate stage
- Cascode stage
- **Differential Pair**
 - Differential pair with resistive load
 - **Current source load**
 - Active current mirror



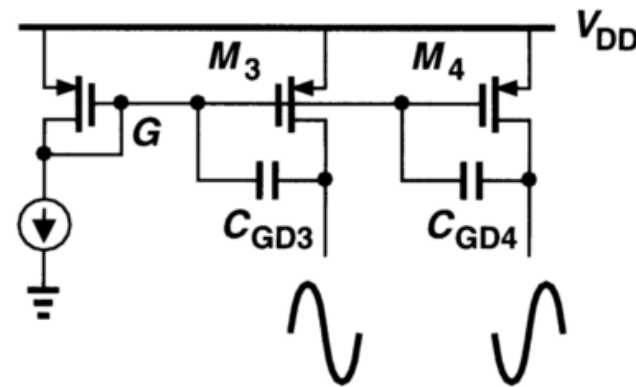
DM frequency response



– Half circuit equivalent

– G is a small-signal ground

• Because of symmetry



– Output pole

$$\omega_{p,out} = \frac{1}{(r_{o1} \parallel r_{o3}) \cdot C_L}$$

• May be the dominant pole



Outline

- General consideration
- Common-source stage
- Source Followers
- Common-Gate stage
- Cascode stage
- **Differential Pair**
 - Differential pair with resistive load
 - Current source load
 - **Active current mirror**
- Gain-bandwidth tradeoff



Qualitative analysis

- The circuits have two paths to the output

- Path 1 (Slow):

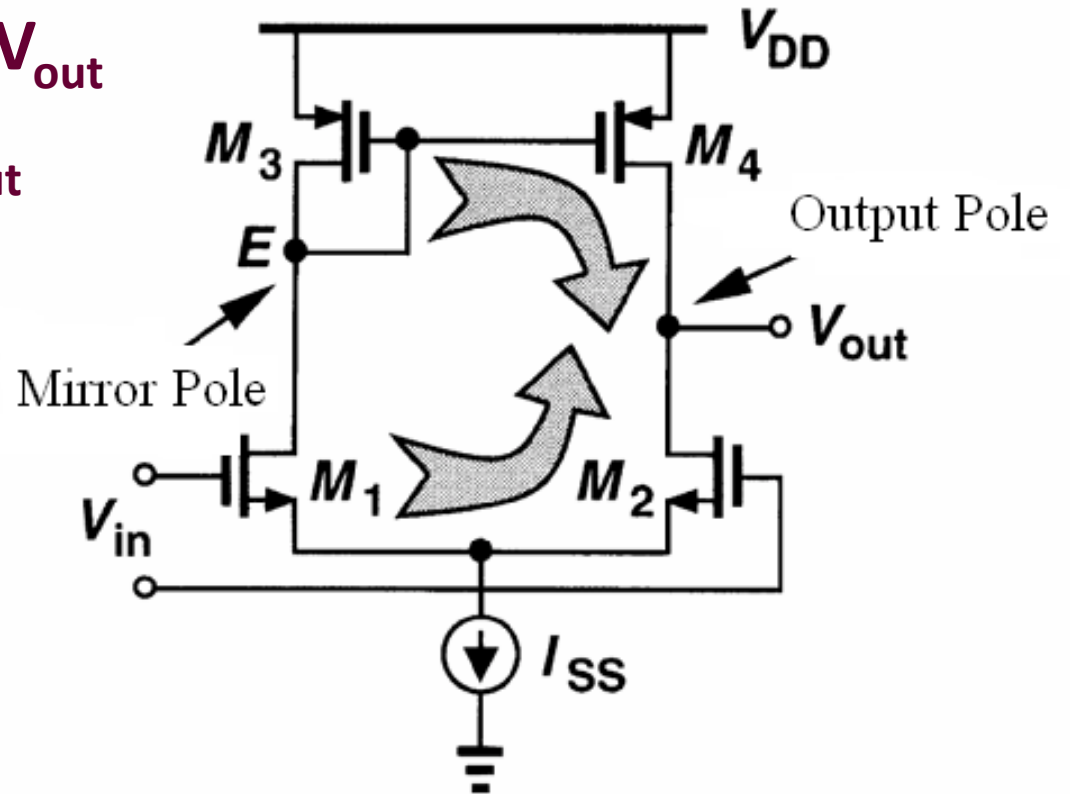
- $M1 \rightarrow M3 \rightarrow M4 \rightarrow V_{out}$

- Poles at E and V_{out}

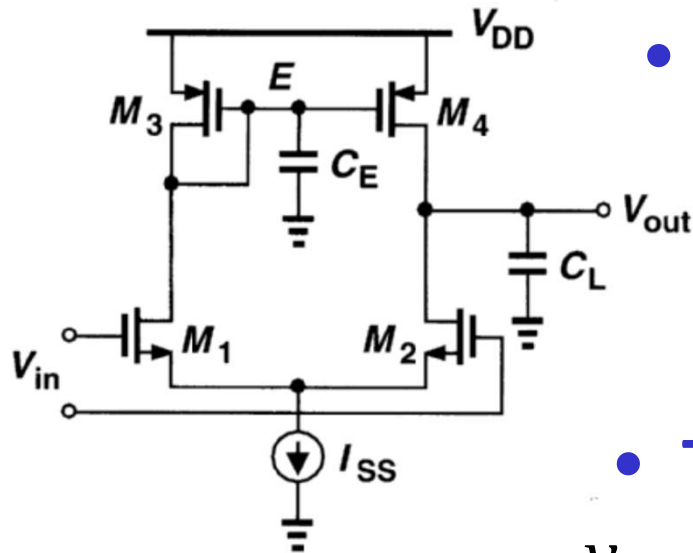
- Path2 (Fast)

- $M1 \rightarrow M2 \rightarrow V_{out}$

- Pole at V_{out}



Determining the poles



- Output pole

- Mirror pole

$$\omega_{p1} \approx \frac{1}{(r_{oN} \parallel r_{oP})C_L} \quad \omega_{p2} \approx \frac{g_{mP}}{C_E}$$

- Transfer function of the fast path

$$V_{out,f} = \frac{V_{in}}{2} A_{v0} \cdot \frac{1}{1 + s / \omega_{p1}} \quad A_{v0} = g_{m1,2} \cdot (r_{oN} \parallel r_{oP})$$

- Transfer function of the slow path

$$V_{out,l} = \frac{V_{in}}{2} A_{v0} \frac{1}{1 + s / \omega_{p1}} \frac{1}{1 + s / \omega_{p2}}$$



Overall transfer function

- Transfer function of the fast path

$$V_{out} = V_{out,f} + V_{out,l}$$

$$= \frac{V_{in}}{2} A_{v0} \cdot \frac{1}{1 + s / \omega_{p1}} + \frac{V_{in}}{2} A_{v0} \cdot \frac{1}{1 + s / \omega_{p1}} \frac{1}{1 + s / \omega_{p2}}$$

$$\longrightarrow A_v = \frac{V_{out}}{V_{in}} = A_{v0} \cdot \frac{1 + s / (2\omega_{p2})}{(1 + s / \omega_{p1})(1 + s / \omega_{p2})}$$

– A LHP zero exists at $2\omega_{p2}$



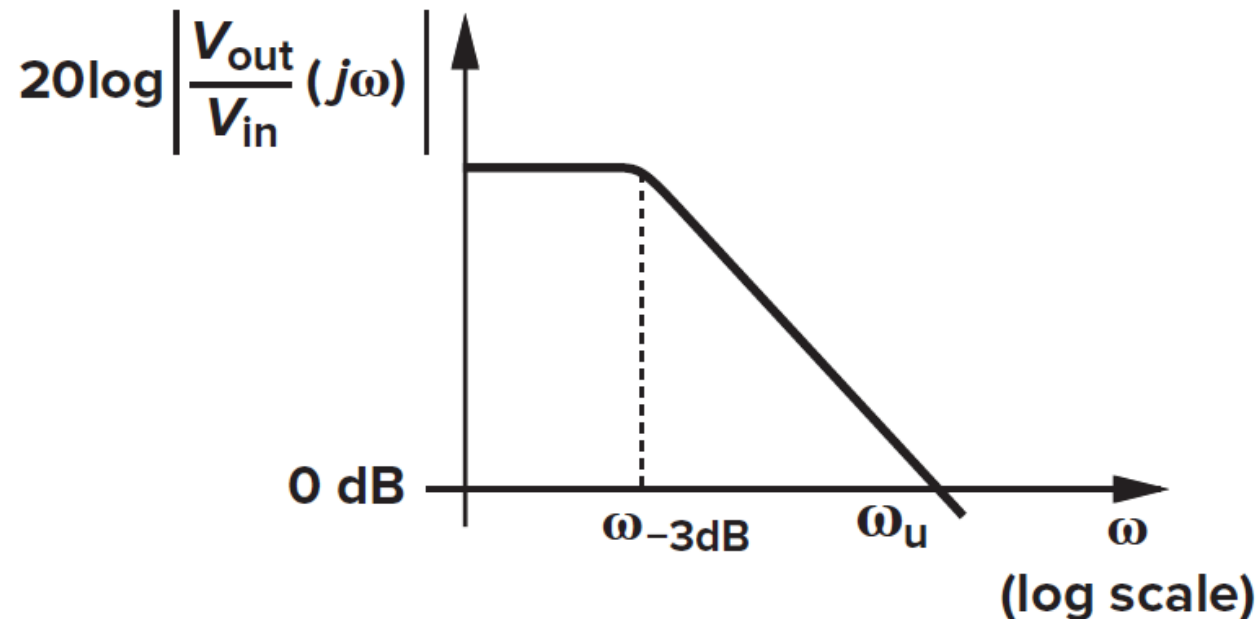
Outline

- General consideration
- Common-source stage
- Source followers
- Common-gate stage
- Cascode stage
- Differential pair
- Gain-bandwidth tradeoff



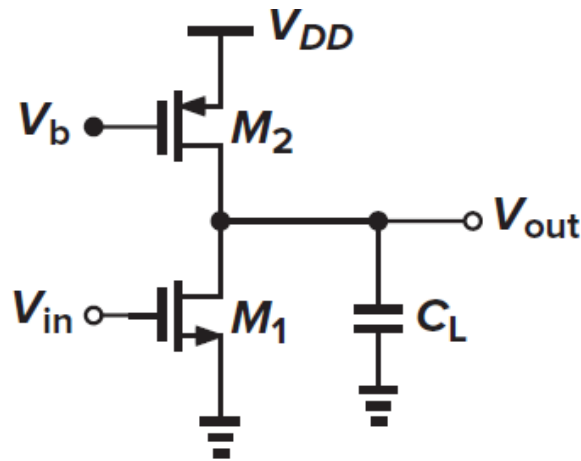
Concept of unit-gain bandwidth

- Besides the -3dB bandwidth, the concept of unit-gain bandwidth (ω_u) is also very important
 - ω_u the frequency at which the gain decreases to 1 (or 0dB)



Single-pole amplifier

- Gain-bandwidth product



$$\begin{aligned} \text{GBW} &= A_0 \omega_p \\ &= g_{m1} (r_{O1} || r_{O2}) \frac{1}{2\pi (r_{O1} || r_{O2}) C_L} \\ &= \frac{g_{m1}}{2\pi C_L} \end{aligned}$$

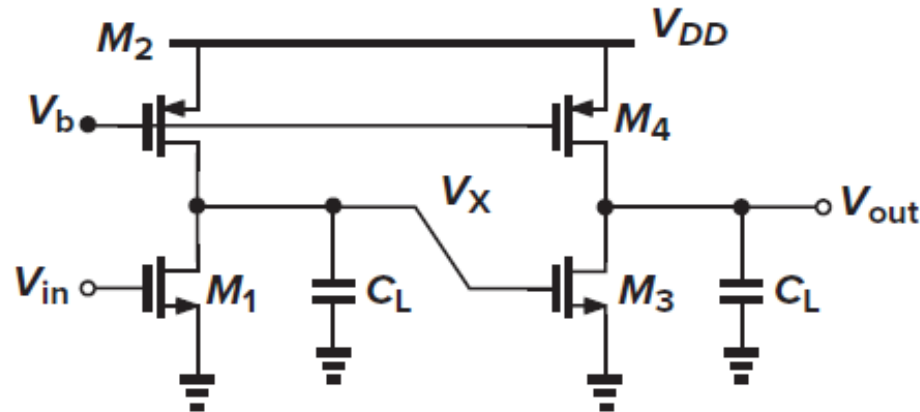
- Its ω_u can be calculated as

$$\frac{A_0}{\sqrt{1 + \left(\frac{\omega_u}{\omega_p}\right)^2}} = 1 \quad \Rightarrow \quad \begin{aligned} \omega_u &= \sqrt{A_0^2 - 1} \omega_p \\ &\approx A_0 \omega_p \end{aligned}$$

$\omega_u \approx \text{GBW} = g_{m1}/C_L$, independent of $r_{\text{out}}!!$



Multipole amplifiers



- For simplicity, neglect other capacitors and assume the 2 stages are identical

$$\frac{V_{out}}{V_{in}} = \frac{A_0^2}{(1 + \frac{s}{\omega_p})^2}$$

$$A_0 = g_{mN}(r_{ON} || r_{OP})$$

$$\omega_p = [(r_{ON} || r_{OP})C_L]^{-1}$$

$$\Rightarrow \frac{A_0^2}{1 + \frac{\omega_{-3dB}^2}{\omega_p^2}} = \frac{A_0^2}{\sqrt{2}} \Rightarrow \omega_{-3dB} = \sqrt{\sqrt{2} - 1} \omega_p \approx 0.64 \omega_p$$

$$GBW = \sqrt{\sqrt{2} - 1} A_0^2 \omega_p$$

GBW is increased, but bandwidth is reduced

For N cascaded stages:

$$\omega_{-3dB} = \sqrt{\sqrt[N]{2} - 1} \omega_p$$



summary

- General consideration
- Common-source stage
 - Miller effect
- Source Followers
- Common-Gate stage
- Cascode stage
- Differential Pair
 - Differential pair with resistive load
 - Current source load
 - Active current mirror

